DEPARTMENT OF INFORMATION SYSTEMS

SYSTEMS DESIGN & DEVELOPMENT



SYSTEMS SPECIFICATION FOR *FUTURE PATH STUDENT ADVISING SYSTEM*

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1. INTRODUCTION

OVERVIEW OF SPECIFICATION

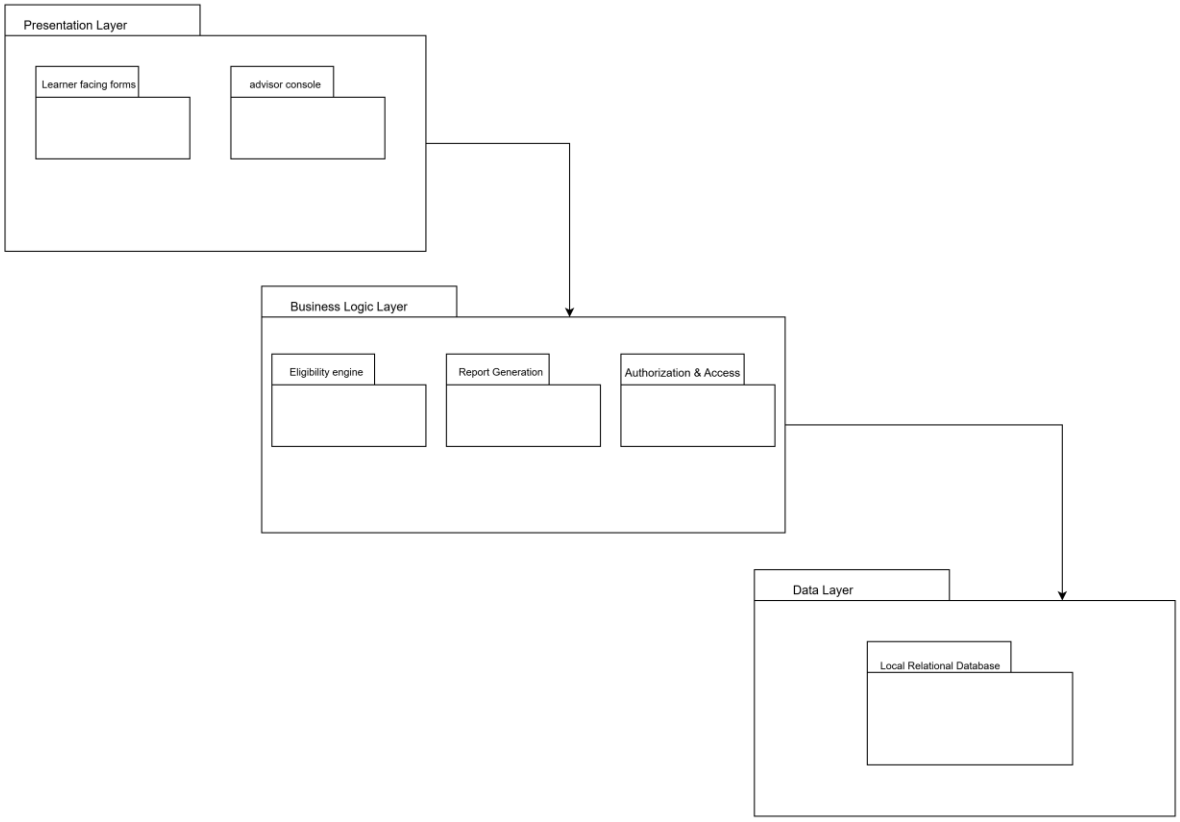
Every year, large numbers of Grade 11 and Grade 12 learners in South Africa must choose a tertiary study path with only fragmented, inconsistent information to guide them, a university website here, a teacher's informal opinion there, outdated pamphlets, or second-hand advice from parents and peers. Because schools often have only one career advisor responsible for large numbers of learners, most learners never get a structured, individualised check of whether their current subjects and marks support the qualifications they are interested in. As documented in the project case study, the result is late discovery of admission mismatches, missed funding opportunities, and study choices that are not well aligned with a learner's actual academic profile or interests.

Future Path addresses this gap directly. It is a standalone, Windows Forms-based advising application that lets a learner capture their grade, subjects, marks and interests, and then uses locally stored eligibility rules and a weighted scoring model to indicate whether they are a Strong Match, Possible Match, Not Currently Eligible, or require More Information for a selected qualification ,along with a plain-language explanation of the outcome, alternative programme suggestions, relevant funding information, and an exportable advisory summary the learner can share with a parent, teacher or career advisor.

CONTEXT & SCOPE OF SYSTEM SPECIFICATION

This design specification builds directly on the objectives and requirements established during the analysis phase: the system must operate entirely offline, using only local business rules and locally stored data, with no external AI services, APIs or cloud platforms involved at any point; it will be built on a three-tier architecture; and its advising outcomes will always be explainable, since Future Path is intended to provide preliminary guidance rather than official admission decisions. These requirements are treated as non-negotiable inputs to every design decision that follows in this document.

OVERVIEW OF SPECIFICATION



Narrative:

The diagram groups Future Path into its three mandated architectural layers: Presentation, Business Logic and Data with dependencies flowing from the Presentation Layer down through the Business Logic Layer to the Data Layer. This specification covers the full vertical slice of the system end-to-end, because the case study requires every learner-facing feature (profile capture, interest questionnaire, eligibility checking, funding lookup, report export) to be underpinned by rule-based logic and locally stored data at every step, a partial-layer design would not satisfy that requirement. Within the Business Logic Layer, the eligibility and scoring engine is treated as the system's critical-path component: it is the dependency that every other business service and presentation form ultimately relies on to produce a usable outcome, and it is the component most directly tied to the project's core value proposition of explainable, rule-based advising. Design and implementation effort is therefore prioritised on this engine and its supporting data structures (subjects, programme requirements, interest categories) before the surrounding presentation forms and secondary services (funding lookup, report export) are finalised.

OVERVIEW OF SPECIFICATION

In-Scope:

● A standalone Windows Forms desktop application using a local relational database, built on a three-tier architecture.

● Learner profile creation, subject and mark capture, and a short interest questionnaire.

● Rule-based eligibility checking and a weighted recommendation engine producing a Strong Match, Possible Match, Not Currently Eligible Match, More Information Needed outcome, with an explanation.

● Suggestion of alternative degree programmes when a learner does not meet the requirements for their first-choice qualification.

● Display of locally stored funding information linked to a selected faculty or degree.

● Export of a learner advisory summary report, plus two further administrative/advisory reports (an aggregate eligibility/interest report and a not-currently-eligible exception report).

● Authentication and role-based access control distinguishing Learner, Career Advisor and System Administrator functionality.

Out of scope for this project:

● Any connection to ChatGPT, Gemini, Copilot, other online AI services, external APIs, cloud platforms or a web server, all advising logic must run locally.

● Making or replacing official university admission decisions; Future Path provides preliminary guidance only.

● Live or real-time funding and admissions data, funding and requirement information is locally stored sample data, not pulled from the internet.

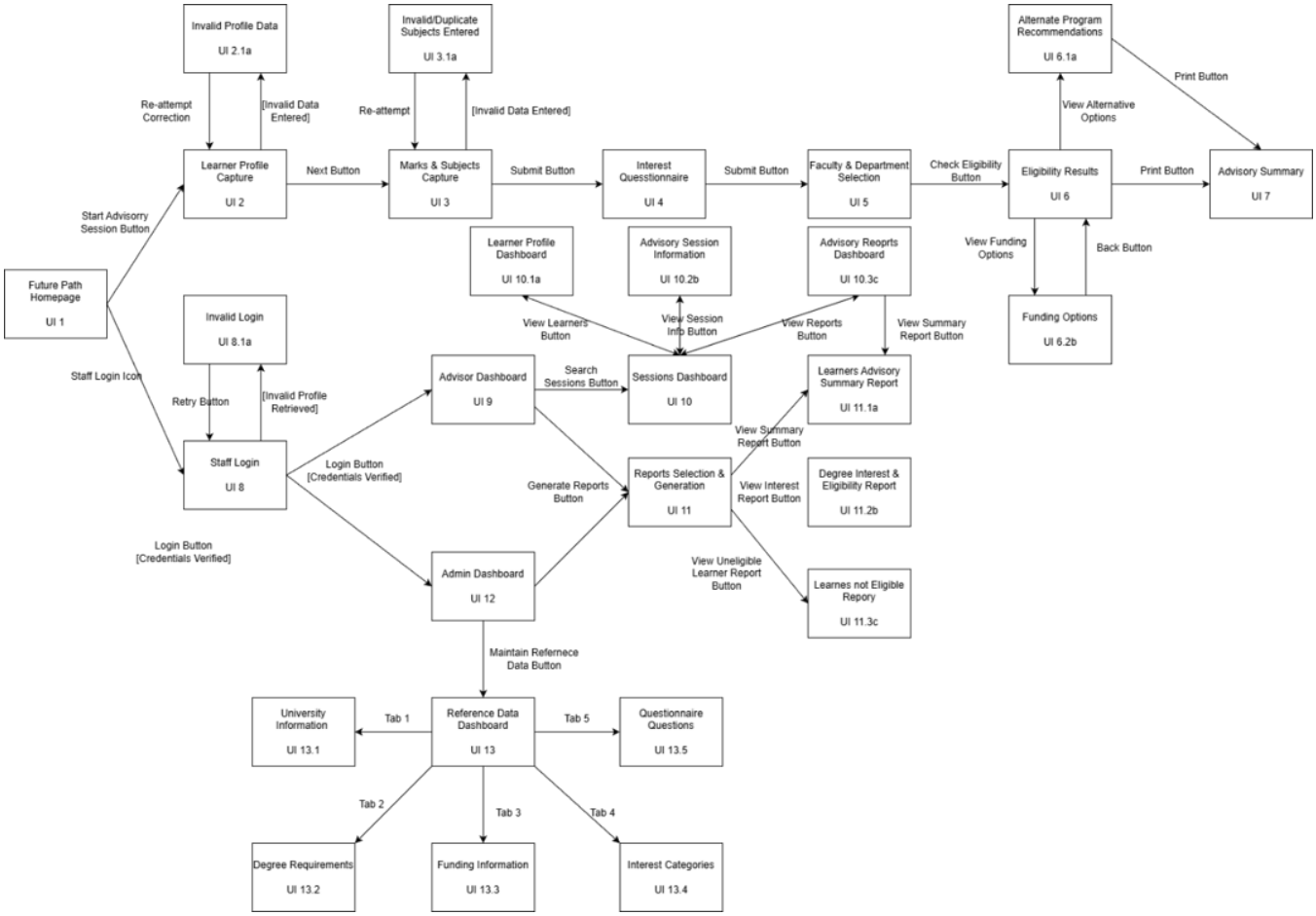
● Submission of funding or university applications on the learner's behalf.

● Support for multiple universities, detailed career pathway mapping beyond the interest questionnaire, parent/teacher login, automated application checklists, closing-date reminders, multilingual support, or natural-language interaction, these are noted as possible future extensions but are not part of the current build.

2. USER INTERFACE & DIALOGUE DESIGN

The user interface for the Future Path system must be designed to ensure that Learners, Career Advisors, and Administrators can navigate the system logically and interact with its major functions. The interface design focuses on form-to-form navigation, specific screens required for profiling, subject and mark capturing, interest questionnaire, eligibility checking, recommendation reporting, and administrator support functions.

INTERFACE FLOW DIAGRAMS



Narrative:

Future Path opens on a single Homepage that splits into two navigating systems, the Learner Advisory System and System Staff Management System. The Management System is an authenticated staff area behind a login that branches by role into Career Advisor and Administrator, with Reports Generation being a shared function.

The Learner side is a six step or optional seven step linear capture system with validation loops that converge on an Eligibility Results screen. From there, Learner can branch into Alternative Program Recommendations and Funding Options that feed into a printable Advisory Summary Report.

The Advisor side is a lookup and view hub that can search through learner profiles and report data and generate necessary reports to improve the system and reference data. The Administrator side mirrors this hub pattern for Reference data maintenance that splits into individual maintenance screens serving specific data categories needed in the system.

Assumptions:

- Faculty & Department Selection has no dedicated error screen as it will be selection based rather than text entry input, so there's less room for invalid input.

- The shared Login condition covers both Advisor and Administrator as their branching will be done internally via the user's role rather than two separate forms.

- Sessions Dashboard is a hub, not a forced sequence. The Advisor can search once and can jump between Learner Profile, Session Info and Reports without researching again.

- Advisory Reports Dashboards shortcuts straight into the Summary Report. The Advisor will be able to get a summary report of all learners' Advisory reports or a selected range of students.

- The Administrator's Maintenance Dashboard groups all categories under one dashboard but keeps its own independent screen with executable controls. There are separate categories for easier maintainability.

SCREEN STANDARDS

The standards below are applied to every screen listed in 2.1. They are fixed before development, and every rule is expressed in terms of the .NET Framework Windows Forms controls and regulations. A screen built to this standard can be designed using Visual Studio without custom painting libraries.

2.2.1) Form Layout and Positioning

- Fixed size Forms, FormBorderStyle = FixedSingle, MaximizeBox = false, StartPosition = CenterScreen, AutoScaleMode = Dpi.
  Three sizes only: wide 1200 × 800 px for capture and record screens, standard 1000 × 720 px for outcome, hub and selection screens. All fit a 1400 × 800 display without scrolling.
- Mandatory zones on every screen. Caption bar reading "Future Path <screen name>", navigation rail docked left, title band carrying the screen name, action band docked bottom with navigation button, and an overall status band for the learners progress.
- Conditional zones. A learner context strip where the screen concerns one learner, an amber notice panel where something is still outstanding, and up to four running-figure tiles where derived values help. Each is hidden entirely rather than shown empty.

- Spacing. Controlled groups of functions and input fields will be positioned about 20px apart from one another.

2.2.2) Navigation Conventions

- The navigation rail is a primary way to move between areas and ongoing processes. “Cancel, Back, and Next” buttons will be placed onto the action band at the bottom of the screen for additional and easier navigation.
- Sequence for learners: UI 2–7 run in a fixed order with the rail showing progress.
- Sequence for staff: Staff screens are destinations reached in any order. The five reference-data maintenance screens are TabPages inside UI 13.
- No dead ends. Every screen offers Back, Cancel or a rail destination. Back always preserves captured data for later completion.

2.2.3) Button Placement and Naming

Position	Label	Style	Behaviour
Furthest Right	Save, Continue, Next, Generate, Print, Check Eligibility	Navy fill, white text	The form’s AcceptButton, so Enter activates it. Disabled while a blocking validation error stands.
Second from Right	Back	White fill, blue border, black text	Returns to previous screen, preserving captured data.
Third from Right	Cancel	White fill, blue border, black text	The form’s CancelButton which abandons the session after a confirmation.
Beside its group	Add Subject, Search, Clear	Amber fill, black text.	Acts on its group, performs validation checks, and does not navigate.

2.2.4) Fonts and Headings

One typeface throughout, Segoe UI, which ships with Windows, so no font needs installing on a laboratory machine and no substitution can break a layout. Consolas is used only for figures, because monospaced digits stay aligned in a column. Hierarchy is carried by size and weight, never by colour, and Semibold is used in place of Bold.

Screen Title – Segoe UI Semibold large (20pt) acting as screen name in the title band.
Section Heading – Segoe UI Semibold medium sized (12pt) in section head bars.
Body – Segoe UI (10pt) for context texts, and grid cells
Field Label – Segoe UI Semibold (12pt) labels above their control
Caption – Segoe UI (8pt) for helper texts and status band
Figures – Consolas (10pt) for marks, scores, dates and identifiers.

2.2.5) Field Labels

- Every input control has a visible label, positioned directly above the control and left-aligned.
- Labels are sentence cases with no trailing colon.
- Units and formats live in the label ie) Mark% and never as a grey placeholder inside the text fields.

2.2.6) Use of Colours

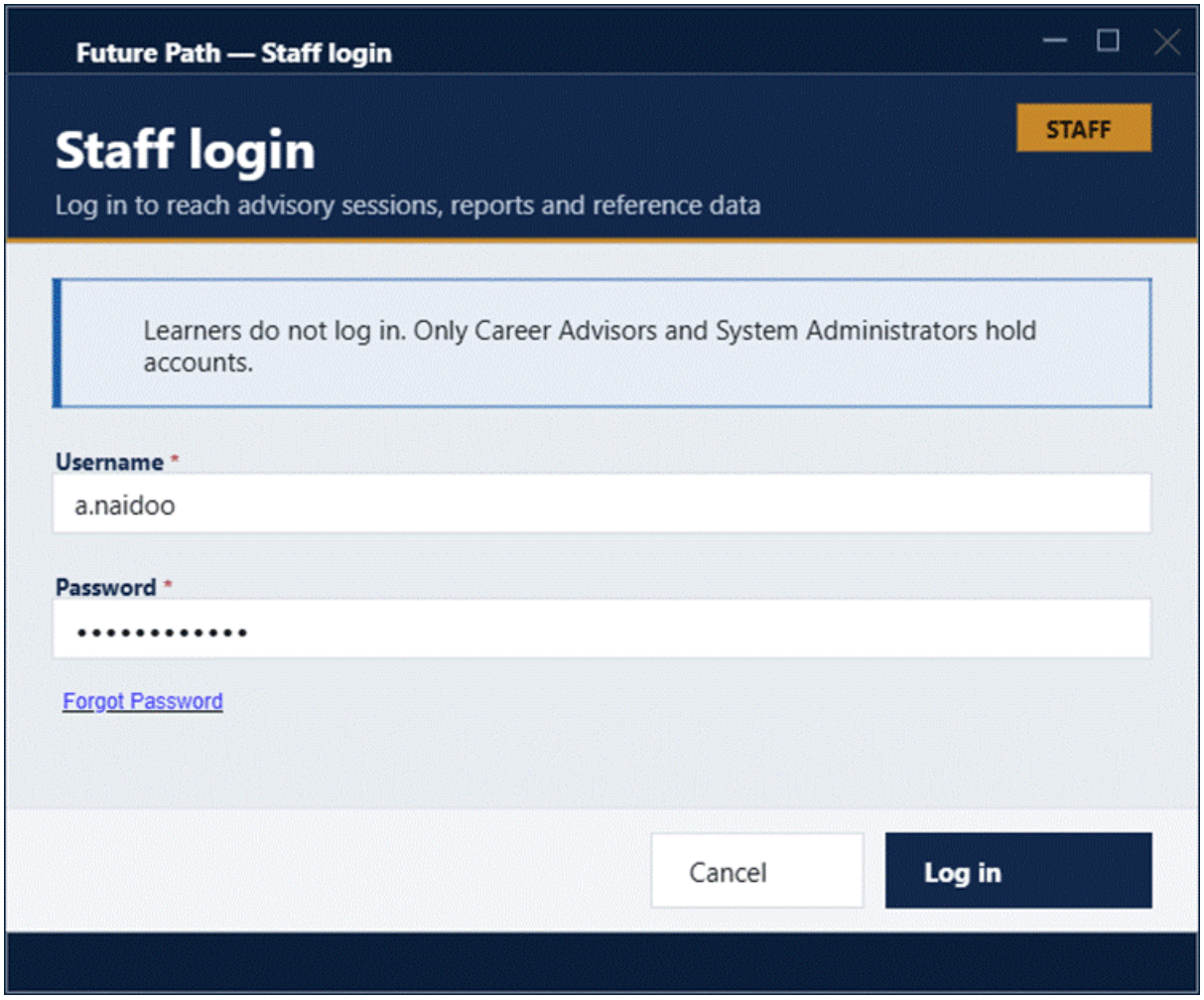
- Navy – Used for structural purposes and the navigation rail, title band, grip column headers and primary buttons.
- Deep Navy – Used to outline the frame to form the outline and status bar.
- Blue – Used to reference the context strip and secondary button borders.
- Blue tint – Used for banding, context strip fills and grid rows.
- Amber – Used to indicated outstanding or incomplete data entry in the notice panel, progress bar and Add button.
- Amber Text – Used to highlight outstanding data.
- Red – Used for invalid input.
- Green – Used to indicate that the requirements have been met and complete. Notice panel changes to notify user can progress.

2.2.7) Error and Confirmation Messages

Class	Presentation and usage
Field Message	Red text with a warning glyph under the control, which gains a red border and tint, raised when the control loses focus with an invalid value.
Validation Summary	Red-bordered box above the controls, one line per offending field, raised when the primary action is activated while any field is invalid.
Validation Dialogue	Modal dialog with a single button, raised when the user tries to leave a capture screen that still holds invalid data.
Status Message	Status band, persistent, signed-in user and role, record or step position, and the count of outstanding validation errors.

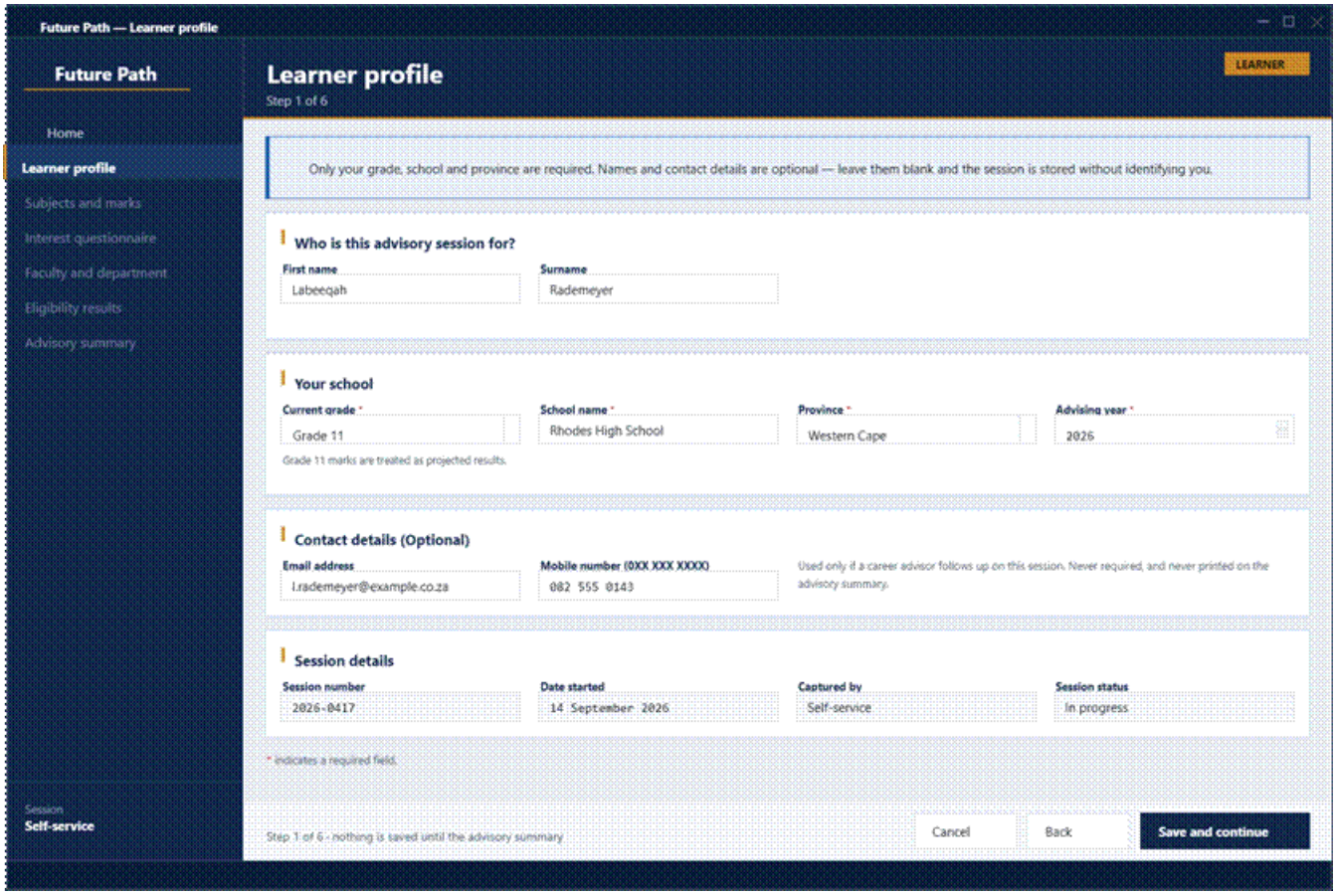
2.3 Detailed Screen Layout

2.3.1) Staff Login (UI 8)



The Staff Login screen uses multiple text area and labels to make up headings and guidance labels. Two text fields are used for user input, where the password text field is masked out for security purposes. Two buttons are also fixed in an action panel for navigation. The user enters only their Username and Password, which is marked as required with a red asterisk. The user's profile will be fetched from the database, and validation will take place upon login attempt. A blank or unrecognised entry highlights the offending label and its text box in red, and the advisory panel changes from blue, (guidance) to a red message naming the field and the correction required.

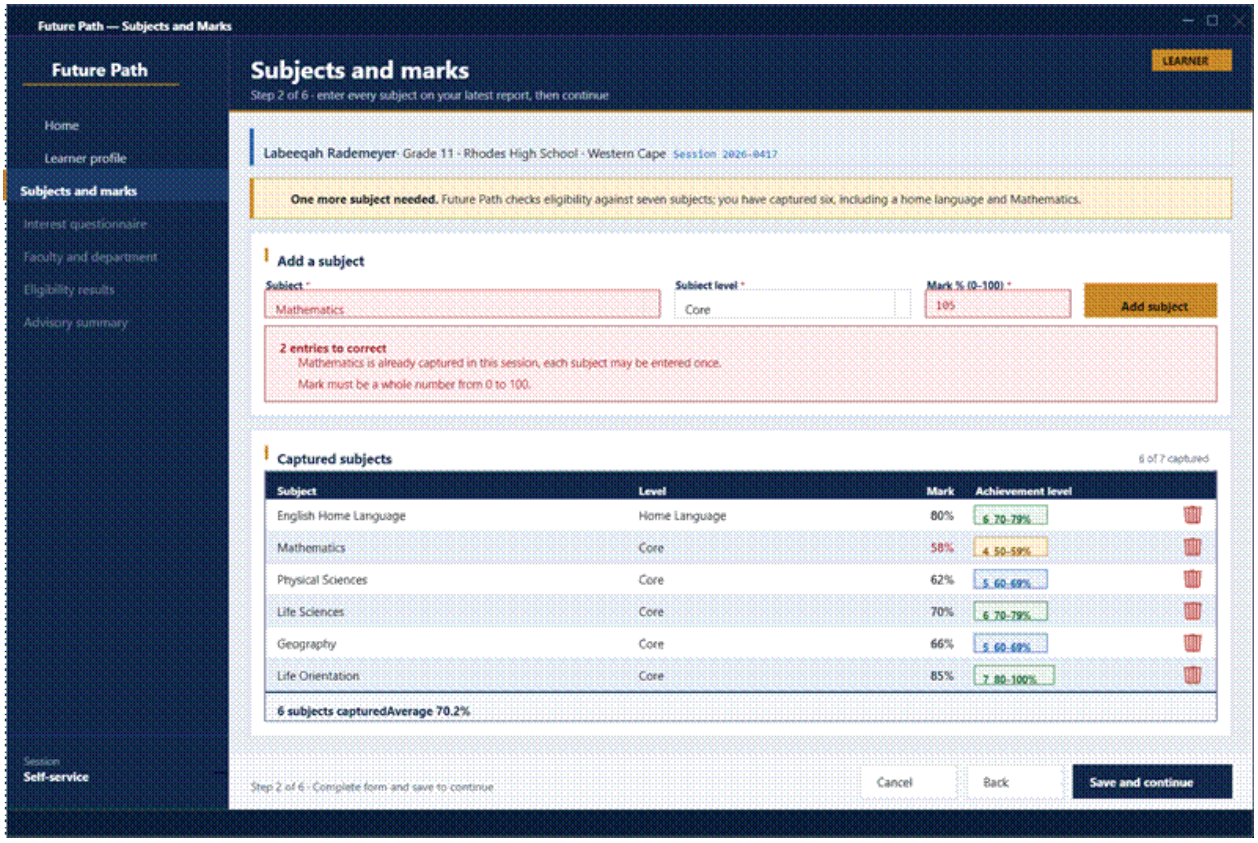
2.3.2) Learner Profile Capture (UI 2)



The Learner Profile screen uses labels, text boxes, combo boxes, a numeric spinner, read-only fields, a blue advisory panel and three command buttons. The four required items, Current grade, School name, Province and Advising year, carry a red asterisk. School name, and Province fields are captured through value-bound controls rather than free text. The year is a numeric control so an invalid value cannot be typed at all.

Email address and Mobile number are optional. The system supplies the four Session details being the session number generated at session start, date started stamped from the system date, and the captured by and status fields are generated upon creation and progress through the form. Validation fires on Save and continue. A missing required value highlights its label and control in red and the advisory panel changes from blue guidance to a red message naming the field and the correction. Cancel and Back exit without writing anything.

2.3.3) Subjects and Marks Capture (UI 3)

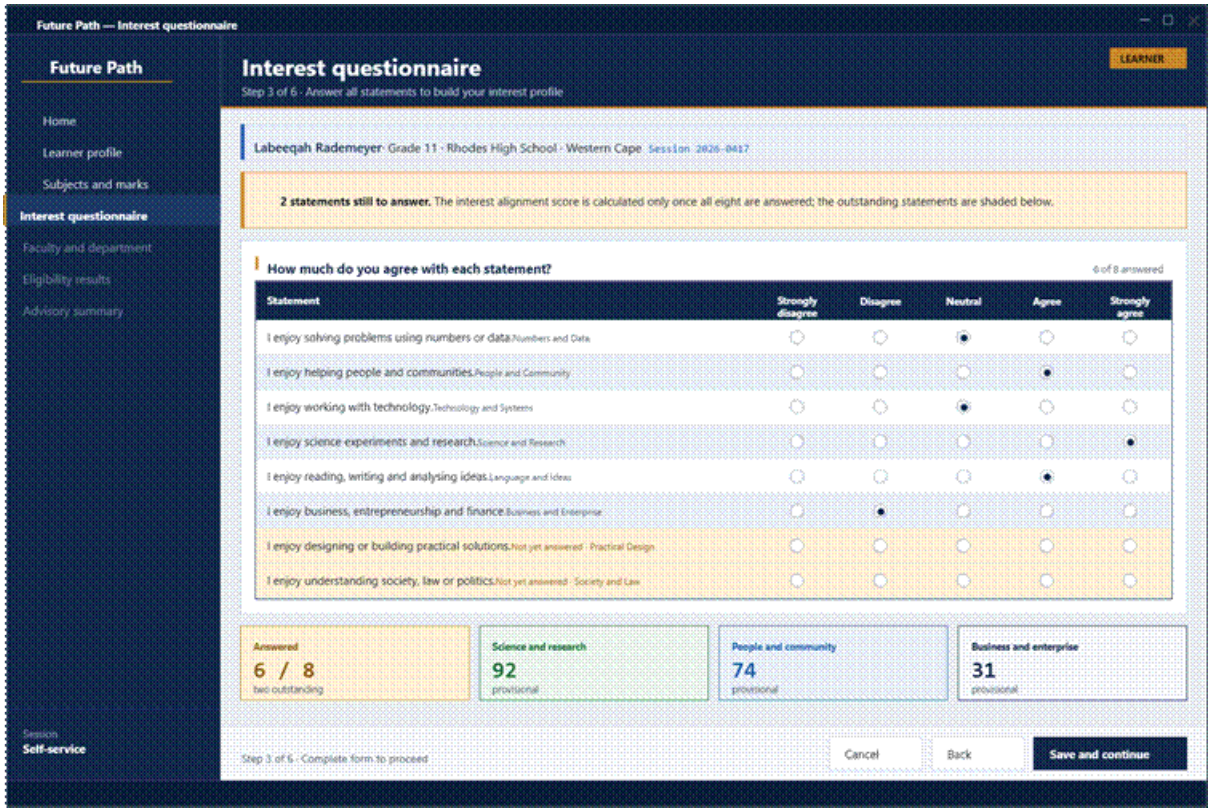


The Subjects and Marks screen use labels, two combo boxes, a numeric control, a data grid, an amber advisory panel, a red validation summary and four command buttons. The user enters three required items per subject, Subject, Subject level and Mark % which is marked with a red asterisk. Subject and Subject level are value-bound combo boxes and Mark % a numeric control constrained to 0–100, so an invalid value cannot be typed.

Add subject commits a row to the grid and the bin icon on a row removes it. There is a minimum subject count of 7 but no maximum, so a learner can add extra Advanced Programme subjects. Every derived value is supplied by the system. The Achievement level, taken from the entered mark against the stored NSC bands, and the captured count and average are both recalculated on each add or delete.

Validation fires on “Add Subject” so a duplicate subject or an out-of-range mark highlights the offending label and control in red, and the validation summary lists every entry to correct, while the amber panel reports how many subjects remain against the stored minimum. Cancel and Back exit without writing anything.

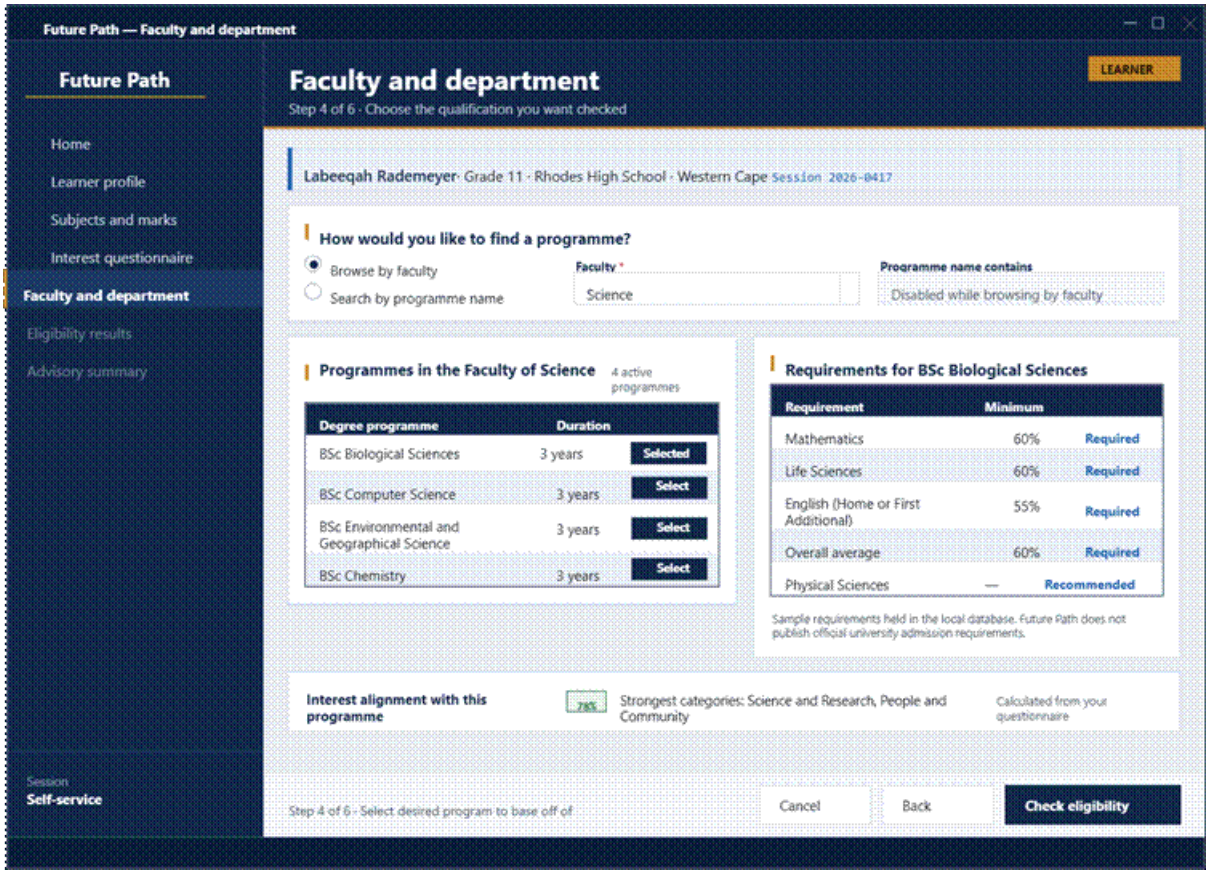
2.3.4) Interest Questionnaire (UI 4)



The Interest Questionnaire screen uses labels, a statement grid carrying eight five-point radio button groups, an amber advisory panel, four running-figure tiles and three command buttons. The user selects one response per statement, from Strongly disagree to Strongly agree, so nothing is typed, and no value on the screen can be invalid.

The system supplies everything else: the eight statements and their display order are read from InterestQuestion and the interest category beneath each from InterestCategory, while the category scores shown in the tiles are calculated from the responses given so far. Unanswered statements are shaded amber and reported in the advisory panel and Save and continue stays disabled until the answered count equals the question count, so amber marks work outstanding rather than an error, and red does not appear on this screen at all. Cancel and Back exit without writing anything.

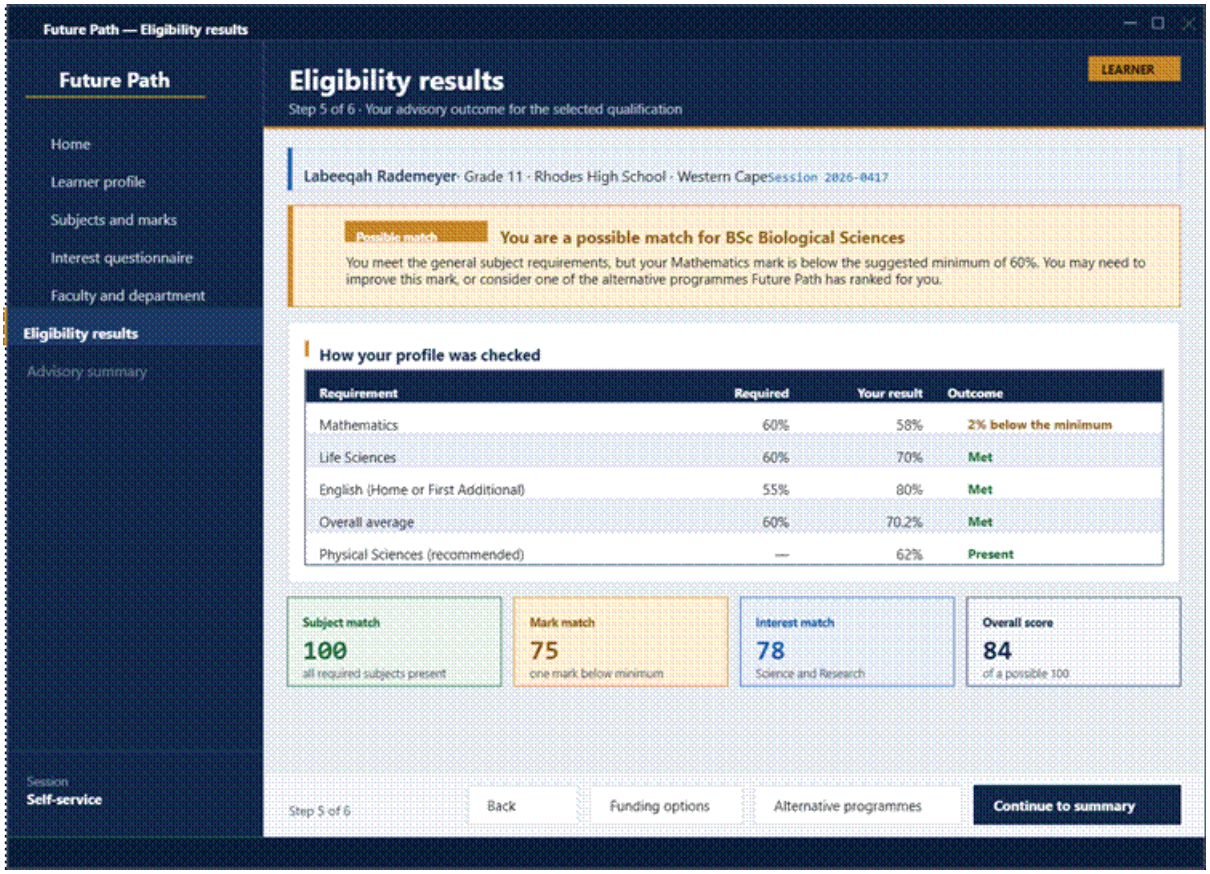
2.3.5) Faculty & Department Selection (UI 5)



The Faculty and Department screen uses labels, a radio pair, a required combo box, a text box shown disabled with its reason stated, two data grids, the first carrying an in-grid Select button column, an interest alignment strip and three command buttons. The user makes three selections: the search mode, the faculty, or one programme via its Select button.

Everything displayed is system-supplied. The faculty combo is bound to the Faculty table and the programme grid. The requirements panel is read from ProgrammeRequirement for the selected programme, and the interest alignment percentage and strongest categories are calculated from the questionnaire. No value can be invalid, and the screen carries no validation dialog. The Check eligibility stays active and repopulates the grids for different selected programs in the department. Cancel and Back exit without writing anything.

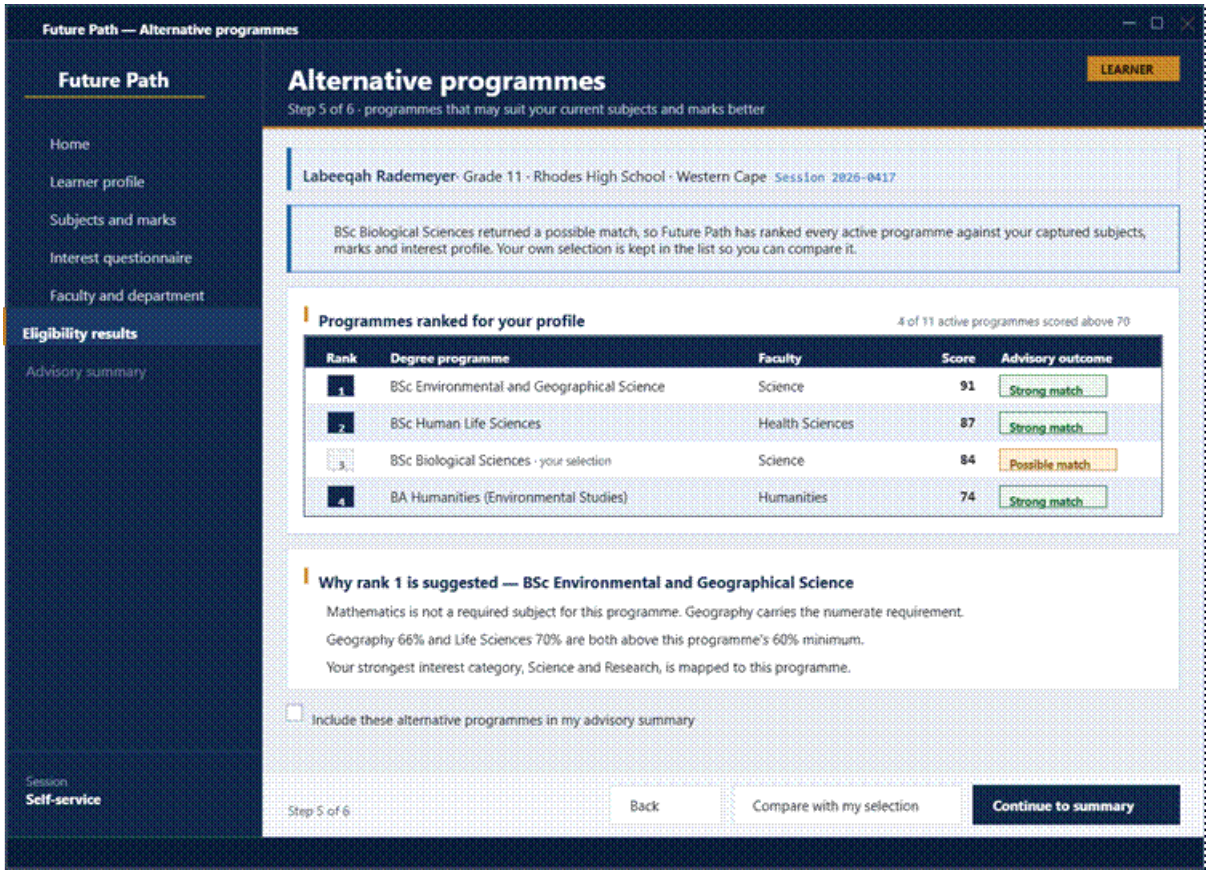
2.3.6) Eligibility Results (UI 6)



The Eligibility Results screen uses labels, a learner context strip, an amber outcome band carrying a status chip, a data grid, four running-figure tiles and four command buttons. Nothing is entered here as it is an outcome screen, and the learner only chooses a forward path or returns to change the programme selected.

Every value shown is produced by the system, the outcome label and its explanation sentence come from the set system rules. With no input there is no validation, only reporting. The band always states why the outcome was reached rather than the label alone, green marks a requirement met and amber a gap, and every state carries its own words so colour is never the only signal. Back, Funding options and Alternative programmes leave the outcome unchanged, and Continue to summary carries it through to UI 7.

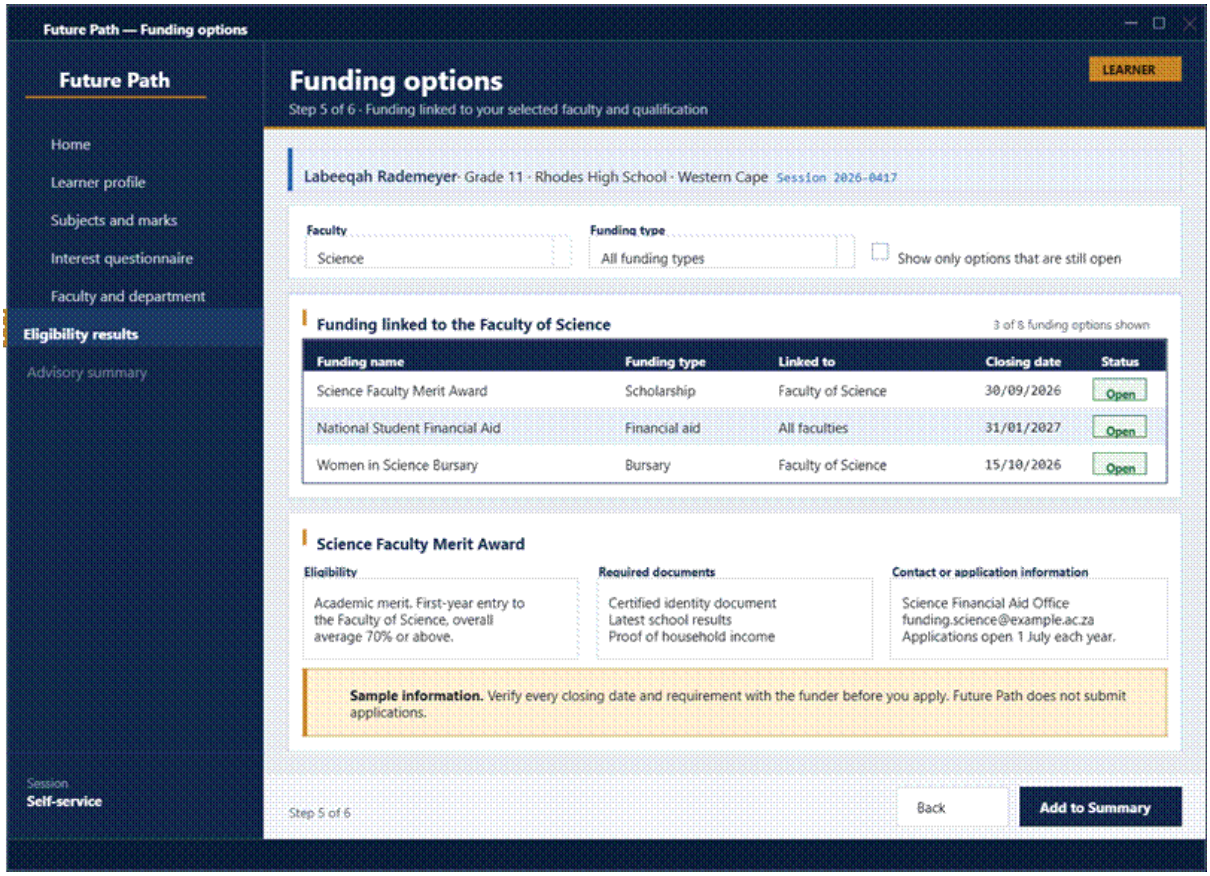
2.3.7) Alternate Programs Wireframe (UI 6.1a)



The Alternative Programmes screen uses labels, a learner context strip, a blue advisory panel, a ranked data grid carrying rank and outcome chips, a reasons panel, a check box and three command buttons.

Nothing is typed, selecting a ranked row rewrites the reasons panel beneath the grid, and the check box is unticked by default, so inclusion is based on the user to decides whether these alternatives are carried into the advisory summary. Everything in the grid is system-produced and derived with internal data. There is no validation, only reporting: every recommendation states its reason rather than rank and score alone, green marks a strong match and amber the learner's own possible match. Back and Compare with my selection change nothing; Continue to summary carries the session to UI 7.

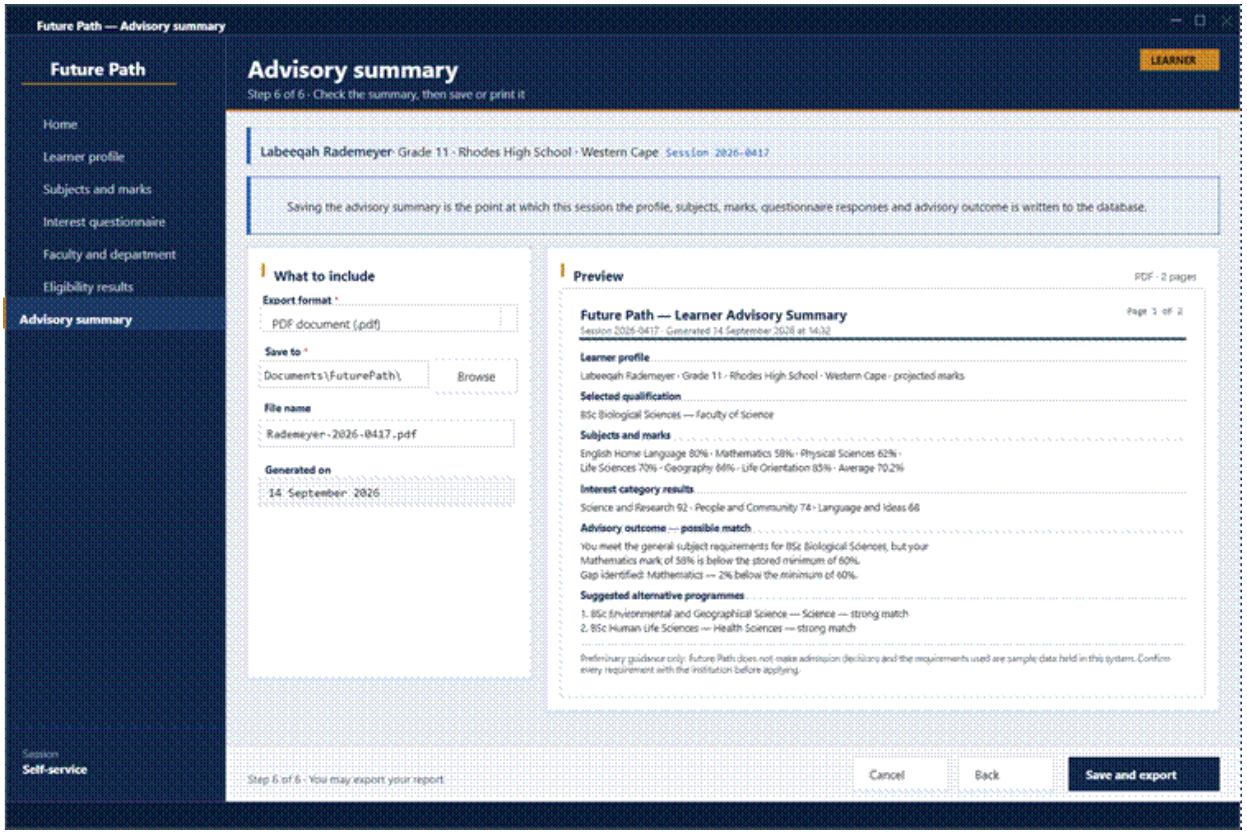
2.3.8) Funding Options (UI 6.2b)



The Funding Options screen uses labels, two combo boxes, a check box, a data grid carrying status chips, a three-field read-only detail panel, an amber advisory panel and two command buttons. Nothing is typed, the learner sets the Faculty and Funding type filters, may limit the list to options still open, and selects a row to load the detail panel.

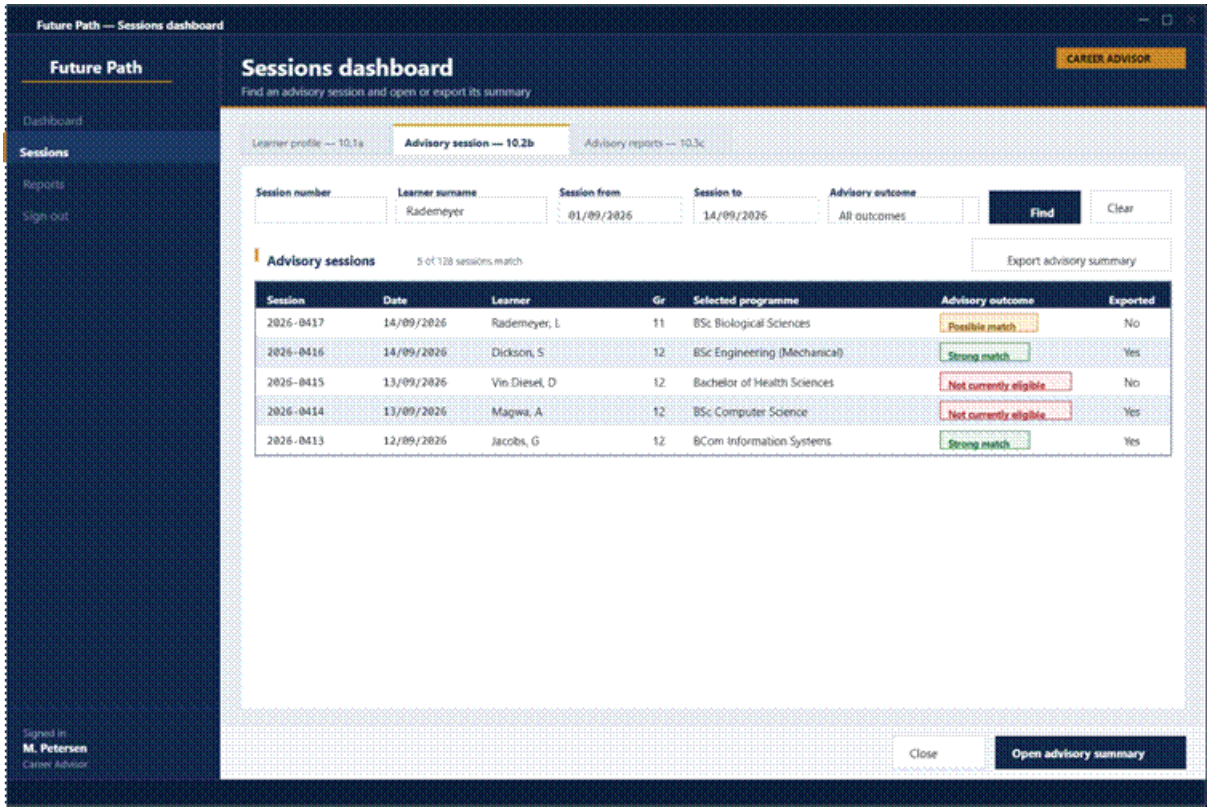
Every value is supplied by the system, and both combos are bound to their reference tables. The rows are read from the database for the faculty chosen on UI 5. Status is derived by comparing each stored closing date with the system date. The count of options shown is derived, and the eligibility, required documents and contact fields are shown as read-only. There is no validation. Back returns to the advisory outcome. Add to Summary carries the funding through to the advisory summary on UI 7.

2.3.9) Advisory Summary (UI 7)



The Advisory Summary screen uses labels, a combo box, two text boxes with a Browse button, a read-only field, a document preview pane, a blue advisory panel and three command buttons. The user sets only the export format and destination folder, both marked required by a red asterisk. Everything else is system-supplied as each line of the preview is assembled from the stored session, the file name is proposed from the learner's surname and session number, the generation date and time are stamped at export, the page count is derived, and the advisory disclaimer is printed automatically. Validation fires on Save and export: an unwritable destination raises a dialog naming the folder rather than a technical error, and success is confirmed in the status band, never as a modal. Save and export commits the session and Cancel and Back exit without writing.

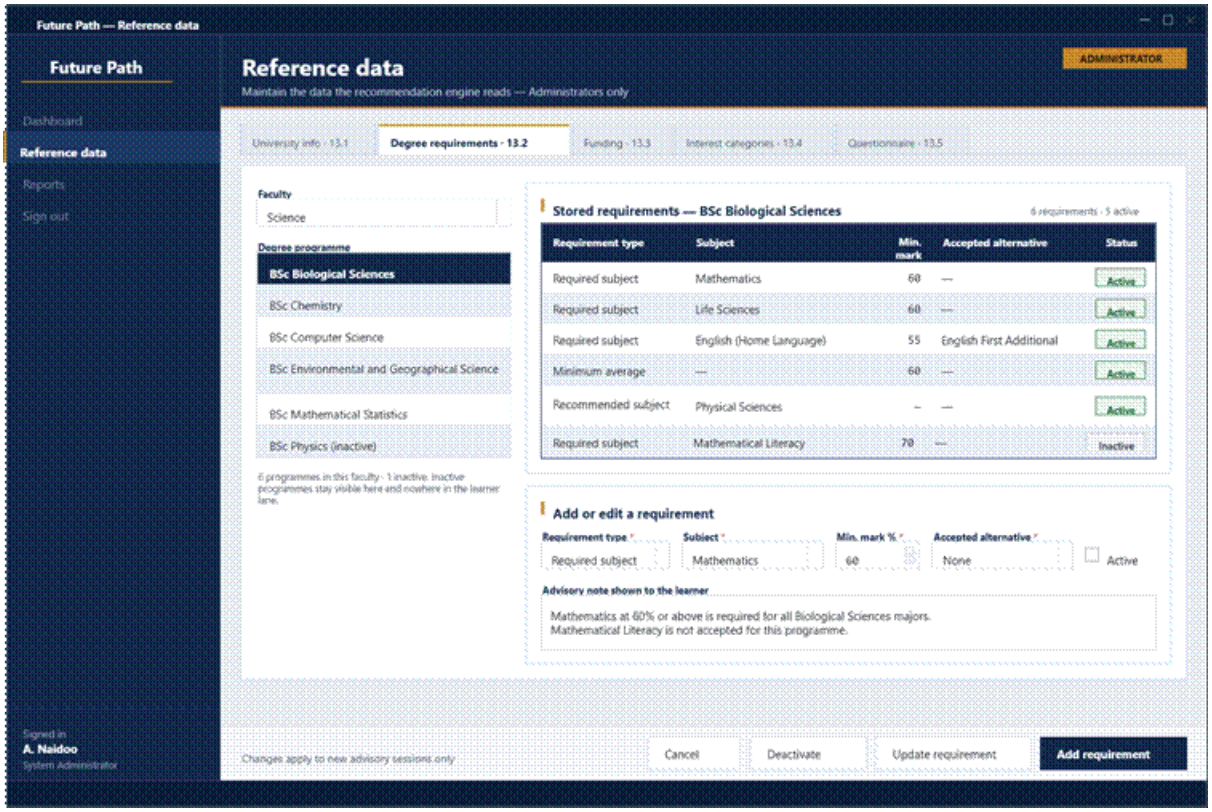
2.3.10) Career Advisor Functionality (UI 10.2b)



The tab uses a tab strip, labelled TextBoxes, DateTimePickers and a ComboBox, a read-only DataGridView with colour-coded outcome chips, command buttons and a status band. The advisor enters only the search criteria, the session number, learner surname, a session date range and an advisory outcome, none of which is required, as any may be left blank to widen the search.

All displayed data is system-retrieved, and the outcome is read back from the stored recommendation result rather than recalculated. On Find, an invalid session number or a from-date later than the to-date highlights the offending label and control in red, with the status band reporting it, turning amber when no sessions match and returning to blue once a valid result set is shown. The remaining buttons capture nothing.

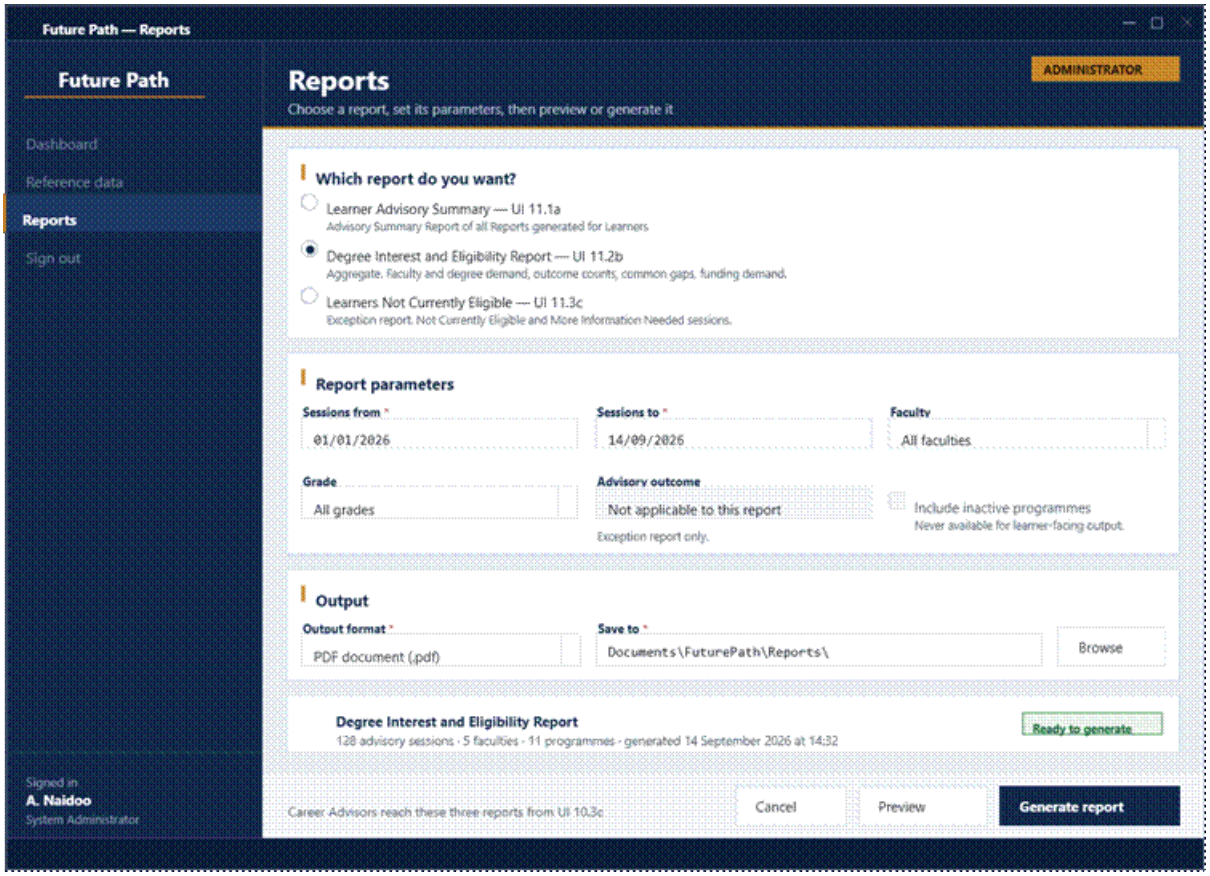
2.3.11) Reference Data Management (UI 13.2)



The tab uses a tab strip, a Faculty ComboBox above a Degree programme ListBox, a read-only DataGridView of stored requirements with colour-coded status chips, a maintenance form of ComboBoxes, a numeric control, a CheckBox and a multi-line TextBox.

The administrator selects a faculty and programme to scope the view, then sets Requirement type, Subject, Min. mark % and Accepted alternative, each marked mandatory by a red asterisk and captured through a value-bound control, so an invalid value cannot be typed. All displayed data is retrieved, the programme list filtered by faculty, the grid rows for program requirements, the lists bound to their reference tables, and the requirement and active counts. Validation takes place on Update requirement or Add requirement which saves mandatory field and highlights a mark outside 0–100 as the offending label and control in red and reports it in the status band. Green marks an active requirement and grey an inactive one. Cancel, Deactivate, Update requirement and Add requirement complete the maintenance cycle.

2.3.12) Report Generation Wireframe (UI 11)

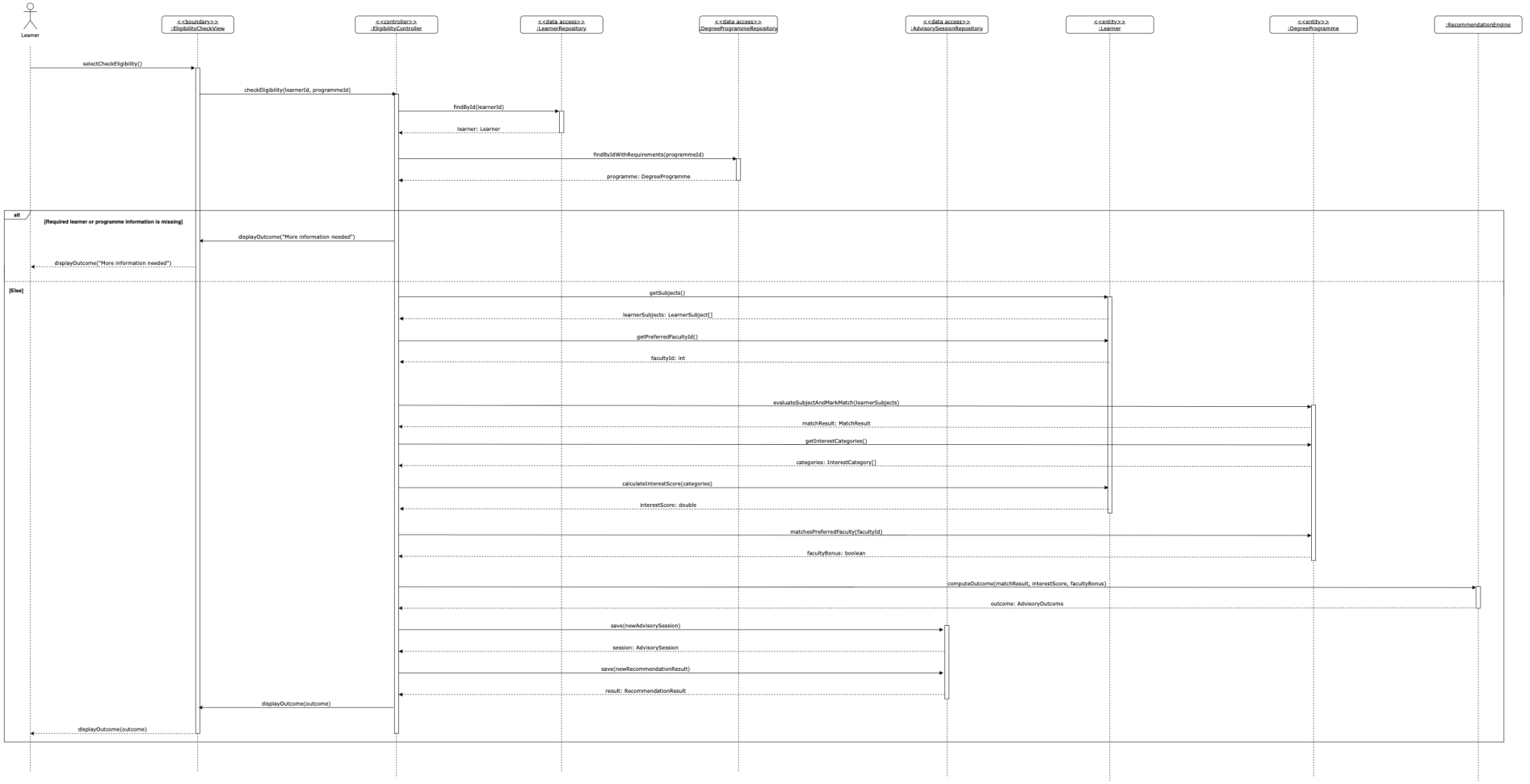


The screen uses a RadioButton group of the three reports, labelled DateTimePickers and ComboBoxes for the parameters, a CheckBox, a TextBox with a Browse button for the destination, a readiness panel and a command band.

The user selects one report, sets Sessions from and Sessions to, Faculty, Grade and Advisory outcome, ticks Include inactive programmes where it applies, and chooses an Output format and Save to path. Sessions from, Sessions to, Output format and Save to are marked mandatory by a red asterisk. The system retrieves and calculates the rest of the data, the three report definitions are read from the report catalogue, the Faculty, Grade and Advisory outcome lists are bound to their reference tables, and the last-generated timestamp is generated from the system. Validation fires on Generate report, where an omitted mandatory field or a from-date later than the to-date highlights the offending label and control in red, parameters that do not apply to the chosen report are disabled, and the readiness panel shows green once every requirement is met. Cancel, Preview and Browse capture nothing.

3. DESIGN SEQUENCE DIAGRAMS

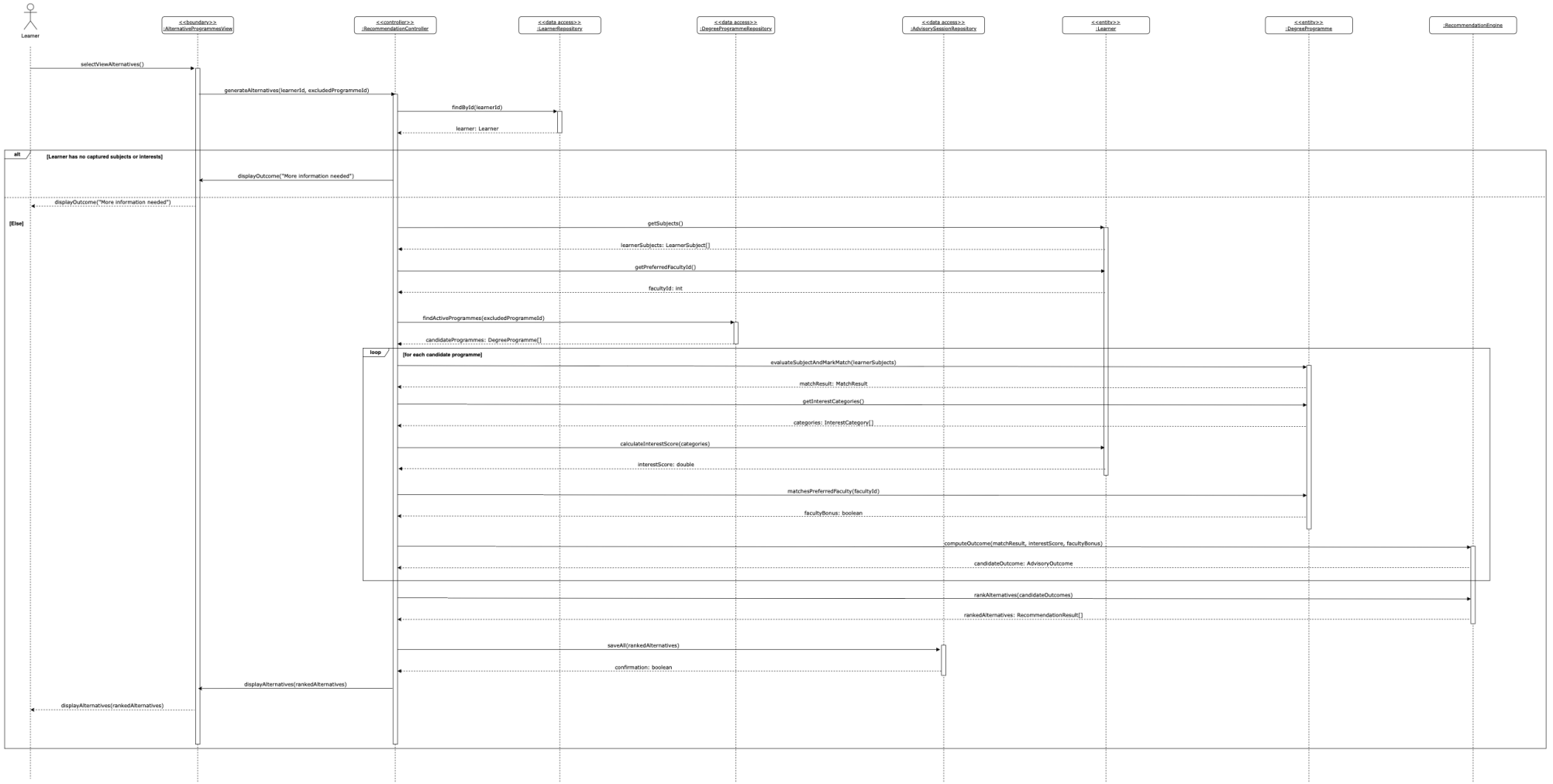
CHECK DEGREE ELIGIBILITY AND GENERATE AN ADVISORY OUTCOME



Narrative:

The process starts when the learner selects the Check Eligibility button from the Faculty and Department Selection screen, this generates the request for the learner ID and programme ID to be sent to the controller. The controller fetches the learner details from the Learner Repository, and the degree programme details from the Degree Programme Repository. If the required learner or programme information is missing, the controller displays a message learner "More Information Needed". Alternatively, if both the details are present, the controller retrieves the subjects captured by the learner and their choice of faculty. Then it calculates the scores required to give out a recommendation. The degree programme determines the match between the subjects and marks and gives out a match result. The interest categories that are associated with the programme are used by the learner to find the interest score, while the degree programme checks whether it matches the learner's preferred faculty and gives out a faculty bonus. All this data is taken and processed by the Recommendation Engine to produce an outcome for the learner. The controller then saves a new advisory session and a new recommendation result, before sending the outcome back to the screen, which displays it to the learner along with its explanation.

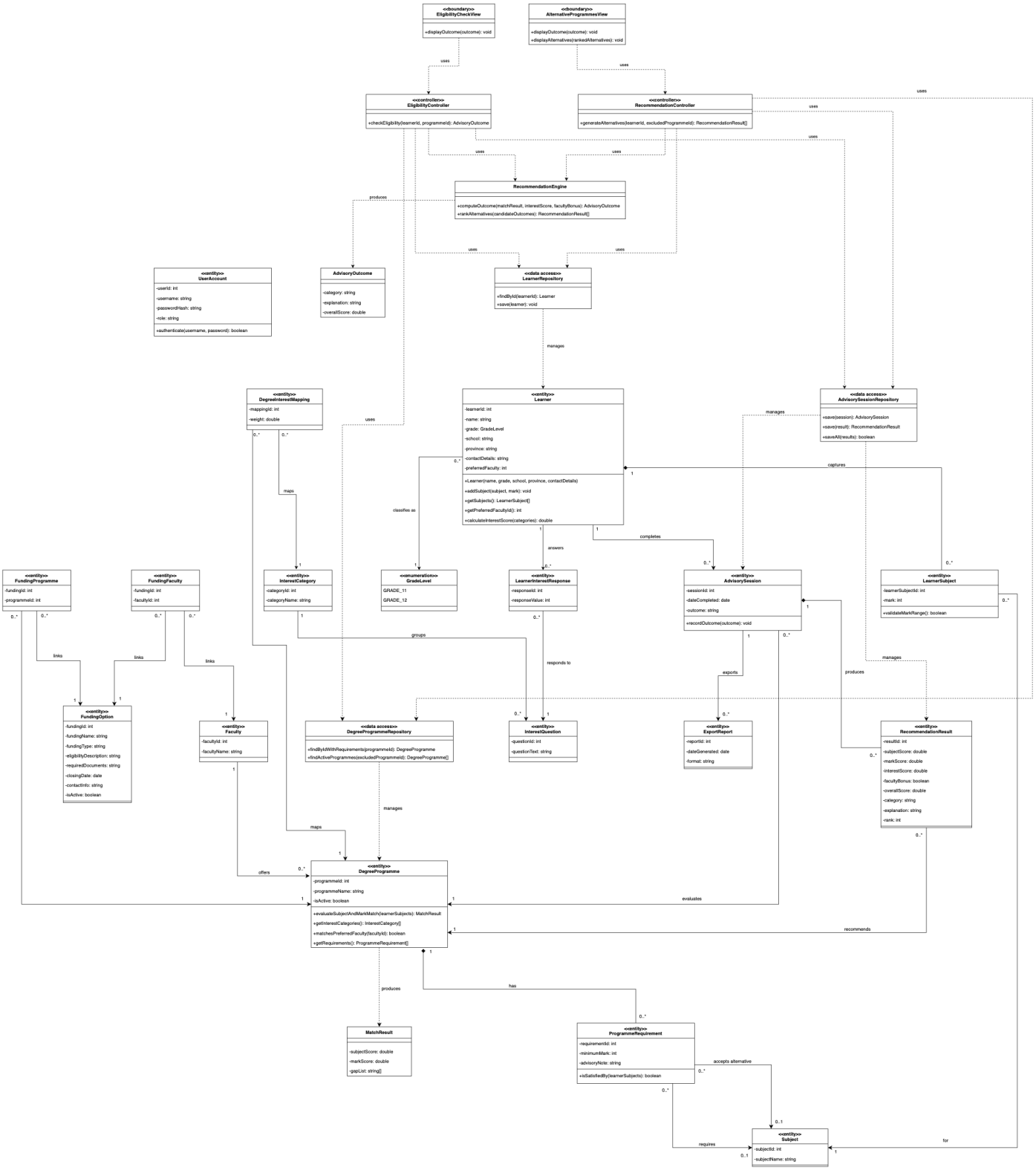
GENERATE ALTERNATIVE DEGREE RECOMMENDATIONS



Narrative:

The sequence diagram begins where the learner chooses to view the alternative programmes in the Eligibility Results screen, this triggers a request for the learner ID and excluded programme ID to be sent to the controller. The controller then retrieves the learner's information from the Learner Repository. If there are missing subjects or interests by the learner, the controller displays a message to the learner "More Information Needed". Alternatively, the controller retrieves the learner's subjects along with the faculty chosen and retrieves the candidate programmes list from the Degree Programme Repository. For each candidate programme, it computes the scores to recommend. The degree programme identifies the match between the subject and marks and gives a match result. The interest categories of the programmes will be used by the learner to compute interest scores. Whereas the degree programme validates whether the programme matches the faculty chosen by the learner and gives a faculty bonus. All these pieces of information are then collected and analysed by the Recommendation Engine to generate a result for each candidate. After scoring all the programmes the Recommendation Engine ranks the candidates and the controller stores the ranked list, which is sent to the learner to display the list of the alternative programme.

4. DESIGN CLASS DIAGRAMS



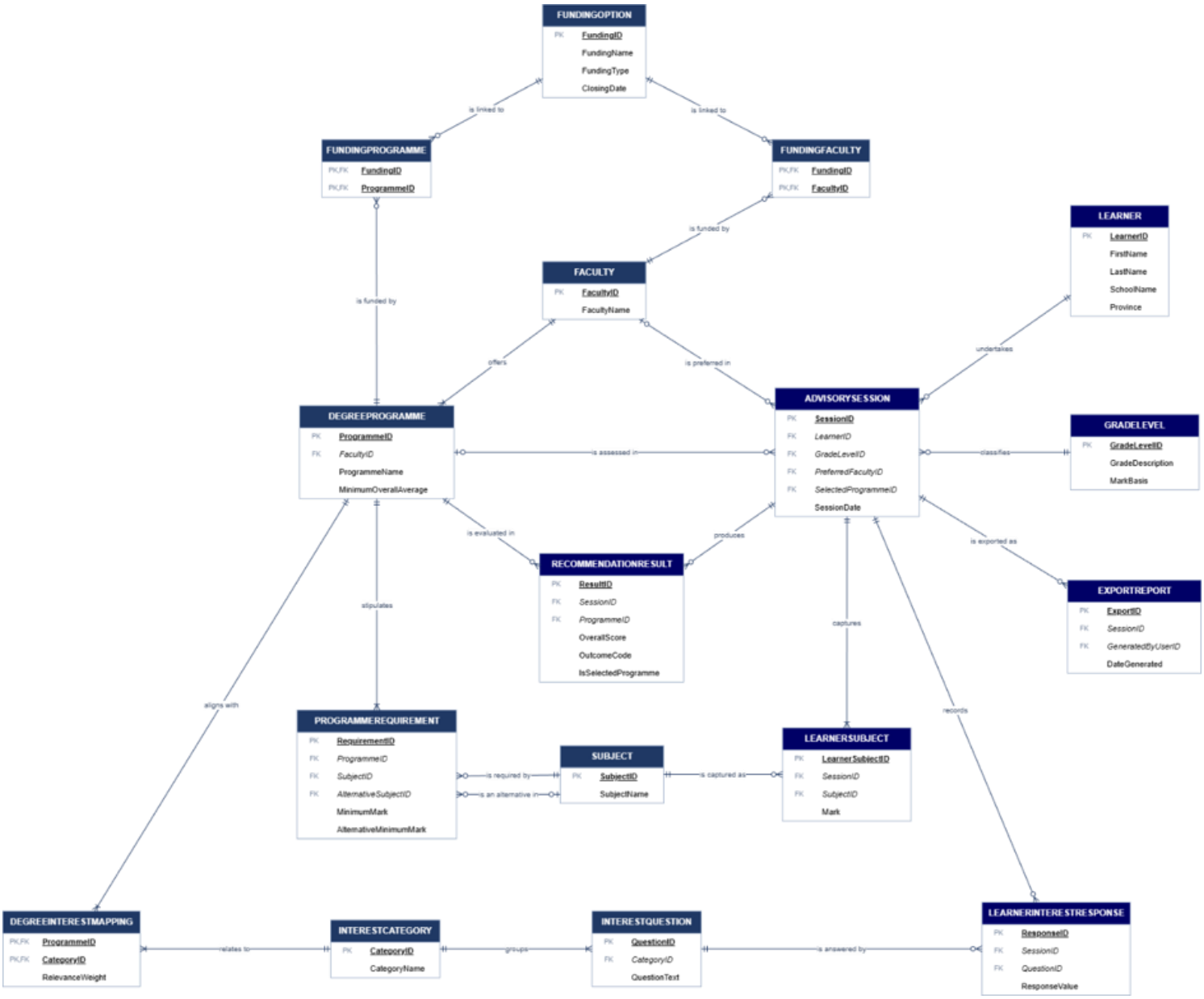
Narrative:

The boundary classes are the screens the user uses to access the system and form the presentation layer. The controller classes form the business logic layer together with the recommendation engine class that manages the scoring logic. The entity classes hold the data used by the system whereas the data access classes are used to store and retrieve the data from the database. These entities consist of the Learner, Degree Programme, Programme Requirement, Advisory Session, and Recommendation Result. These classes handle the profile and subjects of the learners, qualifications, their requirements and the results of the advisory sessions. The two classes, Match Result and Advisory Outcome represent the outcomes for the sequence diagrams. They are both temporary values that are used during the process of generating recommendations. The User Account class has been left disconnected because the learners go through an entire advisory process without logging in.

5. ENTITY RELATIONSHIP DIAGRAM

Narrative:

Each faculty offers one or more-degree programs, and each program stipulates one or more requirements, a required subject, a minimum mark, and optionally an accepted alternative. A learner undertakes zero or more advisory sessions. Each session records the grade level, preferred faculty and selected program, and captures the learner's subject marks and questionnaire responses along with their profile. Each session then produces zero or more recommendation results, one for the selected program, one per alternative, and may be exported as an advisory summary. Funding options reach faculties and programs through two junction tables. Every other relationship is one-to-many, and descriptive values are stored once and referenced by key.



6. REPORT DESIGN

6.1 Learner Advisory Summary Report

The Learner Advisory Summary is the document a learner using the system walks away with at the end of an advisory session once all profile and user data have been captured. They choose to export the report on UI7 at the end of their advisory session, which produces a two-page pdf or docx report. It is generated by the learner themselves, or a Career Advisor and System Administrator can also regenerate the same summary for the specified learner. The report generated is the value itself, an advisory report that gives guidance to students looking to pursue further education on their selected program choice, alternate options, and funding options for their degree.

6.1.1 Detailed Output Requirements

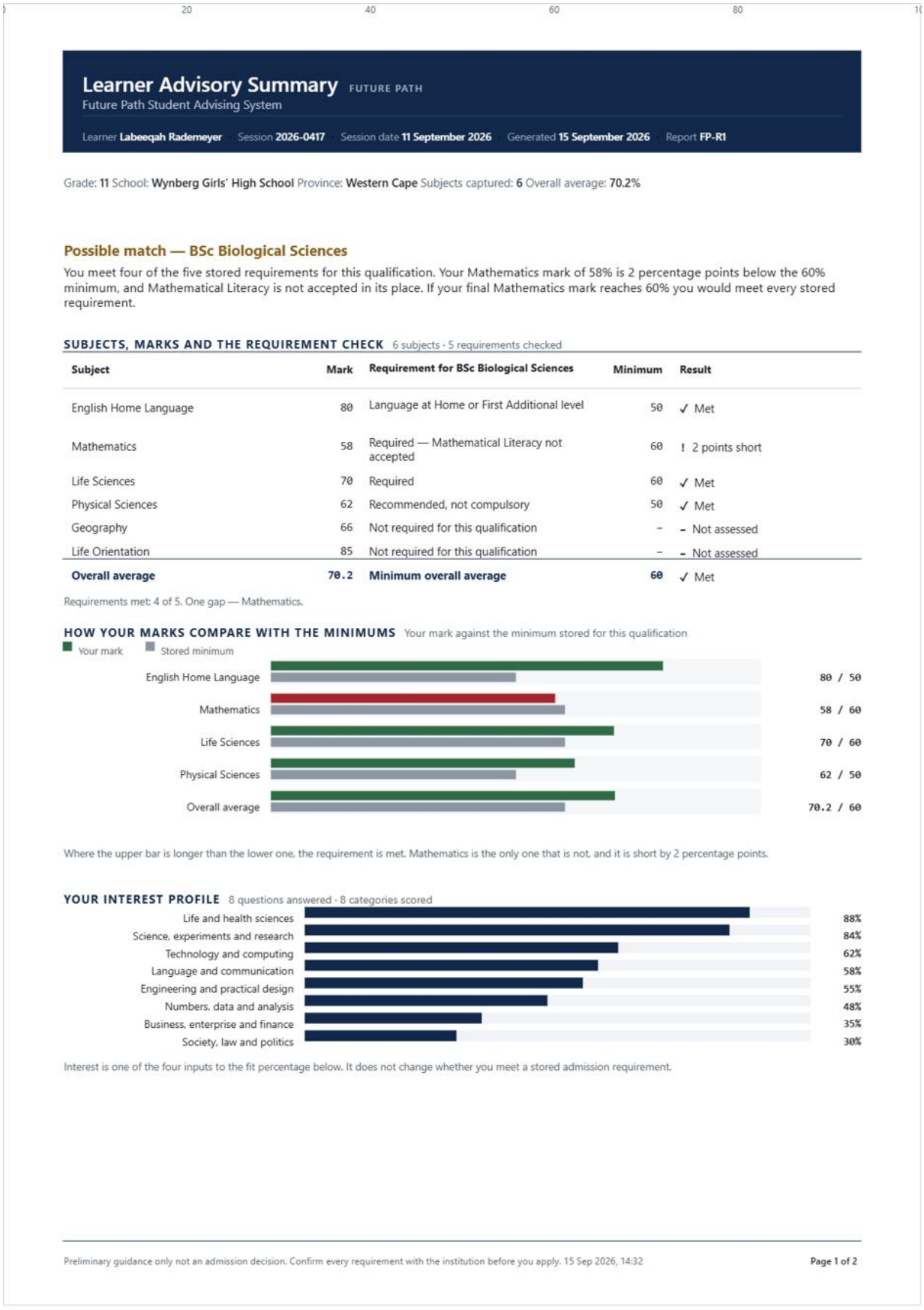
Output type & ID:	Report Type: Detailed Report for single instances, one per advisory session. Report ID: It is a two-paged A4 portrait document carrying the ID: FP-LSR. Report Name: Learner Advisory Summary Report
Report Objectives:	This report aims to give the learner one sharable record of everything the session produced, what they captured, their program selection, the system conclusions and recommendations giving the learner guidance on steps they can take next.
Audience:	The Learners accessing the system are the audience, and the people they choose to share it to being their parents, advisors, or teachers are the secondary audience.
Content:	The fields that are displayed make up the report which consists of a masthead, identity line, advisory outcome, subjects and marks, how those subjects compare with the minimum requirement for each program, an interest questionnaire outcome, alternate program choices and funding options linked to that program. What will be displayed are the following: Report title, learner name, session number and date, report ID, learner's grade, school, province, subjects and corresponding marks and their overall average. Additionally, the program selection and set minimum requirements will be displayed with comparisons to their captured marks, a drafted

	interest questionnaire outcome with fitted categories in charts, alternative program options and funding options the learner can apply for. An overall guidance section will be included
Layout:	Three tables and two charts across two pages. Table 1 and 2 will include 5 columns with supporting data, Table 3 will have 6 columns and supporting data. Only horizontal rules will apply, no box or zebra stripping. Text columns will be left-aligned and numeric columns will be right-aligned in tabular figures. This report will include a grouped bar chart with two bars per subject indicating a visible gap between their current marks and what is required.
Selection:	Only one report per advisory session indicated by the SessionID. All learners using the system will be able to select this report, but requirement rows are restricted to the qualification the learner selected. Alternatives are stored on the recommendation table for qualifications that better suit the learners marks and interest categories, still restricted to the qualifications offered by the institutions that are still active. Funding options only appear if the learner has opted for its inclusion. Invalid or expired programs and funding options aren't added to the report.
Sequence:	Subjects are sorted by subject name in ascending order. Requirements are stored in a requirement order. The mark comparison chart follows as a visual summary of the above information so they can be read together. The interest categories questionnaire output is stored in descending order by score. Alternative programs are displayed next in descending order by overall fit and the funding options display thereafter ascending by closing date, so the nearest deadline appears at the top.
Grouping and Summarization:	There are no set group bands for this report. The report only covers one advisory session, so there is nothing to group by. Summarisation is in four categories:

	- Overall average of the learners' captured marks which is compared against the qualifications minimum overall average. - Count of requirements met for the qualification. - Shortfall in percentage points for each unmet requirement. - Interest percentages per interest category. Fixed percentages on the alternatives table are read from the stored overall score and not recalculated at report time.
Media to be Used:	Electronic document formats like pdf and docx will be used to export the report from the system and saved to a user specified folder. The report will be set to A4 and match the preview, and this report can be printed out as a physical copy.
Frequency, Timing, & Delivery:	The report will be available on demand in real time once per advisory session. On the moment the learner finishes all selection and data capturing, the report can be exported. It may be regenerated at a later stage by a Career Advisor or System Administrator and is never produced on a schedule, in a batch, or for a session that is incomplete.
Distribution:	Reports will be rendered by RDLC from a typed dataset. The report definition holds no queries or business rules so the learner, advisor, or administrator can save the PDF to their local computer, print it and distribute it fellow students, teachers, advisors or parents of the learners. The report is not emailed or transmitted anywhere.
Privacy, Security, and Integrity Constraints:	Access to the learner's own summary is bound to the session in progress, there is no screen where a learner can retrieve another learner's report. The staff access is restricted through authentication and role-based access controls. Their credentials are stored hashed in the database. Every generation writes an Export report row in the database recording session information about the report generated. This is the report's audit trail. Integrity is protected by excluding deactivated qualifications and

	funding options, so a learner is never advised towards something the institution has withdrawn. The report captures its own reference data and each reprint gives rise to new reference data which differs from the original.

6.1.2 Report Layout



0 20 40 60 80 10

Learner Advisory Summary FUTURE PATH

Future Path Student Advising System

Learner **Labeeqah Rademeyer** · Session **2026-0417** · Session date **11 September 2026** · Generated **15 September 2026** · Report **FP-R1**

Grade: **11** School: **Wynberg Girls' High School** Province: **Western Cape** Subjects captured: **6** Overall average: **70.2%**

Possible match — BSc Biological Sciences

You meet four of the five stored requirements for this qualification. Your Mathematics mark of 58% is 2 percentage points below the 60% minimum, and Mathematical Literacy is not accepted in its place. If your final Mathematics mark reaches 60% you would meet every stored requirement.

SUBJECTS, MARKS AND THE REQUIREMENT CHECK 6 subjects · 5 requirements checked

Subject	Mark	Requirement for BSc Biological Sciences	Minimum	Result
English Home Language	80	Language at Home or First Additional level	50	✓ Met
Mathematics	58	Required — Mathematical Literacy not accepted	60	! 2 points short
Life Sciences	70	Required	60	✓ Met
Physical Sciences	62	Recommended, not compulsory	50	✓ Met
Geography	66	Not required for this qualification	–	- Not assessed
Life Orientation	85	Not required for this qualification	–	- Not assessed
Overall average	**70.2**	**Minimum overall average**	**60**	✓ Met

Requirements met: 4 of 5. One gap — Mathematics.

HOW YOUR MARKS COMPARE WITH THE MINIMUMS Your mark against the minimum stored for this qualification

Your mark · Stored minimum

Where the upper bar is longer than the lower one, the requirement is met. Mathematics is the only one that is not, and it is short by 2 percentage points.

YOUR INTEREST PROFILE 8 questions answered · 8 categories scored

Interest is one of the four inputs to the fit percentage below. It does not change whether you meet a stored admission requirement.

Preliminary guidance only not an admission decision. Confirm every requirement with the institution before you apply. 15 Sep 2026, 14:32 **Page 1 of 2**

6.2 Degree Interest & Eligibility Report

The Degree Interest and Eligibility Report is the aggregated counterpart to the Learner Advisory Summary Report. This report reads every completed advisory session in a chosen period and reports what learners are trying to register for and how many qualify for those programs. It is generated by a Career Advisor or System Administrator on demand. It's valuable as it turns individual advisory work into a picture of demand and eligibility across the institution. This reports what qualifications attract most students, in which faculties they are in, and their proportions and which subjects are responsible for requirements not being met.

6.2.1 Detailed Output Requirements

Output type & ID:	Report Type: Summary Report aggregating many advisory sessions into one output generated once per requested period. Report ID: It is a three-paged A4 landscape document carrying the ID: FP-DIE Report Name: Degree Interest and Eligibility Report
Report Objectives:	This report aims to show the volume and distribution of the advisory activity over a selected period to report the eligibility rate achieved for each faculty and each degree program. This report allows users to identify which degree program attracts the most learner interest and to expose the specific subjects and marks that most often prevent learners from enrolling to their desired program.
Audience:	The Career Advisors and System Administrators use this report. Career Advisors use it to see where their guidance is most needed across the learner population. System Administrators use it to judge whether the stored program requirements, interest questions and funding options are producing sensible outcomes in the report.
Content:	Page 1 carries the report masthead, the parameter line showing exactly which sessions were included, a table of advisory outcomes by faculty with an eligibility rate per faculty and total rows. A pie chart summarizes the different outcomes across the whole period as well. Page 2 carries a table of every active degree program selected in the period showing the selections and the eligibility outcomes, which is followed by 5 pie charts showing the outcomes per faculty and

	program. Page 3 carries a bar chart of the required subjects not taken, a table of subjects taken but scored below the minimum requirements with an average shortfall, and a table of funding categories. Every page carries a footer with the report name, generation timestamp and page number.
Layout:	A single navy masthead band at the top of each page holding the report name, period selected, and generation details. Each page will be a landscape A4 layout so that all table sections and figures fit on the page. Section headings are numbered 1 – 8 continuously across all pages. All text is left-aligned, and all numbers are right-aligned in their respected columns. Charts are placed directly beneath the table they explain and every chart is followed by a short note stating what is the takeaway from the diagrams.
Selection:	The user supplies a start date and an end date, and may optionally filter by faculty, by learner grade, and by advisory outcome. Only advisory sessions that were completed and committed within the set period are included. Deactivated degree programs and funding options are excluded by default. Every parameter uses is printed on page 1 so that the user can clearly see what was counted and what was not counted.
Sequence	The faculty table stored in the database is sorted by the number of sessions in descending order so the busiest faculty appears first. The Degree Program table is sorted by the number of selections in descending order so that the most requested qualification appears first. The subject gap bar chart, the mark shortfall table and the funding category table are each sorted by frequency in descending order, putting the largest problem or the highest demand at the top of every list. The five faculty pie charts follow the same order as the faculty table so that the two can be read against each other.

Grouping and Summarization	This report groups advisory sessions first by faculty and then by degree program. Within each group the four advisory outcomes are counted and an eligibility rate is calculated as, strong matches + possible matches, expressed as a percentage of that group's sessions. Both tables close with a totals row summarizing the period in one line. The subject mark and analysis groups items by subject, and the total of this table will exceed the number of sessions as multiple sessions have the same subject entries.
Media to be Used	The report is displayed on screen in the report viewer where the user can page through it and zoom on specifics. The report can be exported to document formats like pdf and excel saved to a user specified folder. The report may be printed as an A4 landscape paper.
Frequency, Timing, and Delivery	The report is generated on demand and is not scheduled or produced automatically. It is typically generated at the end of a term, semester or academic year. Usually ahead of a university application deadline. Generation is immediate and the report is produced from data already stored in the system.
Distribution	The report is delivered only to authenticated users of the Future Path system, being the Career Advisors and System Administrators. Career Advisors uses the information for targeted group reviews and learner support. If exported, it is written to a specific folder within the users computer in a user specified format in PDF or Excel format. The system does not email, upload or transmit the report anywhere outside system testing and internal use to improve the system and its reference data. Any further distribution is an independent act carried out by the user.
Privacy, Security, and Integrity Constraints	Only authenticated Career Advisors and System Administrators may generate this report. Learners have no access to it. The report is aggregated and contains no learner names or profile data. It describes institutional advisory performance and remains restricted to those two roles. The report is mainly read only and generation cannot alter stored session information. The report captures its own

	reference data and each reprint gives rise to new reference data because the period information is different. The disclaimer printed on the final page states that these outcomes are preliminary rule-based guidance and not admission decisions.

6.2.2 Report Layout



Degree Interest and Eligibility Report FUTURE PATH
Advisory activity across all learners

Period **1 January – 15 September 2026** · Generated by **A. Naidoo, System Administrator** · Generated **15 September 2026, 14:32** · Sessions in scope **128** · Report **FP-DIE**

Sessions from: **01/01/2026** – **15/09/2026** Faculty: **All faculties** Grade: **All grades** Outcome: **All outcomes** Inactive qualifications: **Excluded**

Advisory sessions	Strong match	Possible match	Not currently eligible	More information needed	Eligibility rate
128	41	52	29	6	72.7%
96 learners advised	32.0%	40.6%	22.7%	4.7%	strong + possible

ADVISORY OUTCOMES BY FACULTY Sessions sorted by sessions

Faculty	Sessions	Strong match	Possible match	Not currently eligible	More information needed	Eligibility rate
Faculty of Commerce	34	15	13	5	1	82.4%
Faculty of Science	31	10	12	7	2	71.0%
Faculty of Engineering and the Built Environment	24	7	10	6	1	70.8%
Faculty of Health Sciences	21	4	8	8	1	57.1%
Faculty of Humanities	18	5	9	3	1	77.8%
All faculties	**128**	**41**	**52**	**29**	**6**	**72.7%**

ADVISORY OUTCOMES ACROSS THE PERIOD All 128 sessions, as parts of one whole

Strong match	41	32.0%
Possible match	52	40.6%
Not currently eligible	29	22.7%
More information needed	6	4.7%

Just under three sessions in four end in a strong or possible match.

Aggregated advisory guidance not an admission record. Counts reflect advisory sessions completed in the selected period only.15 Sep 2026, 14:32 Page 1 of 3



Degree Interest and Eligibility Report 1 January – 15 September 2026 FUTURE PATH

QUALIFICATIONS SELECTED All 11 active qualifications · sorted by selections

#	Degree programme	Faculty	Selections	Strong match	Possible match	Not currently eligible	More information needed
1	BCom Information Systems	Commerce	19	9	7	3	0
2	BCom Accounting	Commerce	15	6	6	2	1
3	BSc Computer Science	Science	13	4	4	4	1
4	BSc Engineering (Mechanical Engineering)	Engineering and the Built Env.	13	4	6	3	0
5	Bachelor of Health Sciences	Health Sciences	12	3	5	4	0
6	BSc Engineering (Electrical Engineering)	Engineering and the Built Env.	11	3	4	3	1
7	BSc Biological Sciences	Science	11	4	5	2	0
8	BA Law	Humanities	10	3	5	1	1
9	MBChB	Health Sciences	9	1	3	4	1
10	BSocSc Politics and Governance	Humanities	8	2	4	2	0
11	BSc Mathematical Statistics	Science	7	2	3	1	1
	All 11 active qualifications	**5 faculties**	**128**	**41**	**52**	**29**	**6**

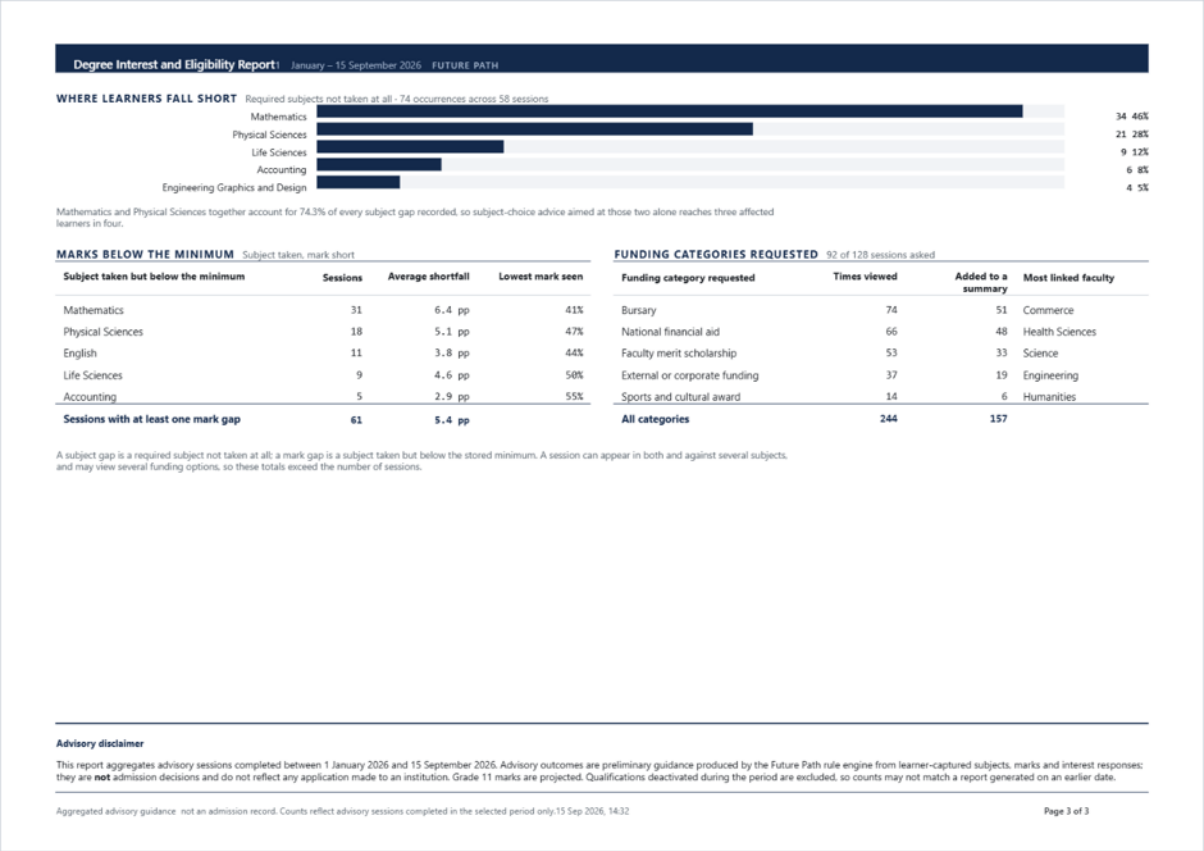
ADVISORY OUTCOMES WITHIN EACH FACULTY The same four outcomes, faculty by faculty

Strong match · Possible match · Not currently eligible · More information needed

Commerce 34 sessions · **Science** 31 sessions · **Engineering and the Built Environment** 24 sessions · **Health Sciences** 21 sessions · **Humanities** 18 sessions

Every pie uses the same slice order, so the red wedge sits in the same place on each one. Health Sciences is the outlier with 8 of its 21 sessions end not currently eligible, against 5 of 34 in Commerce. Read the pies for shape and the table above for the counts behind them.

Aggregated advisory guidance not an admission record. Counts reflect advisory sessions completed in the selected period only.15 Sep 2026, 14:32 Page 2 of 3



Degree Interest and Eligibility Report January – 15 September 2026 FUTURE PATH

WHERE LEARNERS FALL SHORT Required subjects not taken at all - 74 occurrences across 58 sessions

Mathematics and Physical Sciences together account for 74.3% of every subject gap recorded, so subject-choice advice aimed at those two alone reaches three affected learners in four.

MARKS BELOW THE MINIMUM Subject taken, mark short

Subject taken but below the minimum	Sessions	Average shortfall	Lowest mark seen
Mathematics	31	6.4 pp	41%
Physical Sciences	18	5.1 pp	47%
English	11	3.8 pp	44%
Life Sciences	9	4.6 pp	50%
Accounting	5	2.9 pp	55%
Sessions with at least one mark gap	**61**	**5.4 pp**	

FUNDING CATEGORIES REQUESTED 92 of 128 sessions asked

Funding category requested	Times viewed	Added to a summary	Most linked faculty
Bursary	74	51	Commerce
National financial aid	66	48	Health Sciences
Faculty merit scholarship	53	33	Science
External or corporate funding	37	19	Engineering
Sports and cultural award	14	6	Humanities
All categories	**244**	**157**	

A subject gap is a required subject not taken at all; a mark gap is a subject taken but below the stored minimum. A session can appear in both and against several subjects, and may view several funding options, so these totals exceed the number of sessions.

Advisory disclaimer

This report aggregates advisory sessions completed between 1 January 2026 and 15 September 2026. Advisory outcomes are preliminary guidance produced by the Future Path rule engine from learner-captured subjects, marks and interest responses; they are **not** admission decisions and do not reflect any application made to an institution. Grade 11 marks are projected. Qualifications deactivated during the period are excluded, so counts may not match a report generated on an earlier date.

Aggregated advisory guidance not an admission record. Counts reflect advisory sessions completed in the selected period only.15 Sep 2026, 14:32 Page 3 of 3

6.3 Learners Not Currently Eligible Report

The Learners Not Currently Eligible Report is an exception report. It only reports on what needs attention, specifically the advisory sessions in a period that did not end in a match either because the learner did not meet one or more of the requirements for the qualification, they selected or because the session was left incomplete. It is generated by the Career Advisor and System Administrator typically at the start of a follow-up cycle and it names every affected learner, the qualification they were interested in, the reason it could not be recommended and the alternative programs if they were suggested by the system. This report tells the advisor who to contact next and what issues to address.

6.3.1 Detailed Output Requirements

Output type & ID:	Report Type: Exception Report listing only the advisory sessions that fall outside the expected outcome. Report ID: FP-LER Report Name: Learners Not Currently Eligible Report
Report Objectives:	To identify every learner whose advisory session did not result in a match and to present them as an actionable follow-up list. This report states the specific reason each learner did not qualify so that the

	advisor can act on it. It names the program alternatives for each affected learner so that the advisor has something constructive to offer when reaching out.
Audience:	The Career Advisors and System Administrators use this report. Career Advisors use it to do their follow-up work. System Administrators use it to confirm that the stored requirements are producing reasonable outcomes and not excluding learners in error. It is not produced to any learner.
Content:	Page 1 carries the report masthead, a fact line summarizing the sessions in scope, the number and percentage of learners affected. A bar chart of the reasons recorded across the list and a pie chart of which faculties carry the follow-ups. Beneath these are the exceptions grouped by faculty with their session number, date, learner name, grade, school, qualification selected, the outcome and recommended alternative. Page 2 continues listing the remaining faculties and loses with a total row and disclaimer.
Layout:	A single navy masthead band at the top of each page holding the report name, and generation details. Each page will be a landscape A4 layout so that all table sections and figures fit on the page. The two charts sit side by side on page 1 above the listings, so the shape of exception students is outlined. Each faculty is listed along with each learner with their exceptions alongside. these group in tables across the whole report and very faculty listed.
Selection:	The user supplies a start date and an end date which can be optionally filtered by faculty and by learner grade. On the sessions whose advisory outcome is “Not Currently Eligible” or “More Information Needed” are included in this report. The parameters used are printed on page 1 alongside the counts so that the reader can see how many exceptions are drawn from all the sessions recorded.
Sequence	Rows are sorted by faculty first so that an advisor responsible for a faculty can work with a single block of the report. Within each faculty

	they are sorted by outcome, placing those who are “Not Eligible” first. Within each outcome they are sorted by session date with the newest first so the most recent session is more visible as the learner could still be deciding on what to pursue.
Grouping and Summarization	The listing is grouped by faculty, and each group heading summarises that faculty with a count of its exceptions and the split between the two outcomes. The listing closes with a report total row summarising all faculties.
Media to be Used	The report is displayed on screen in the report viewer where the user can page through it and zoom on specifics. The report can be exported to document formats like pdf and docx saved to a user specified folder. The report may be printed as an A4 landscape paper.
Frequency, Timing, and Delivery	The report is generated on demand and is not scheduled or produced automatically. It is typically generated at the end of a term, semester or academic year. Usually ahead of a university application deadline. Generation is immediate and the report is produced from data already stored in the system.
Distribution	The report is delivered only to authenticated users of the Future Path system, being the Career Advisors and System Administrators. Career Advisors uses the information for targeted group reviews and follow ups. If exported, it is written to a specific folder within the users computer in a user specified format in PDF or Excel format. The system does not email, upload or transmit the report anywhere outside system testing and internal use to improve the system and its reference data.
Privacy, Security, and Integrity Constraints	Only authenticated Career Advisors and System Administrators may generate this report. Learners have no access to it. The restriction is printed in the fact line on page 1 so that anyone holding a printed copy can see it is not for general circulation. The report is read only and generating it cannot alter any stored session or requirement. different.

	The report captures its own reference data, and each reprint gives rise to new reference data which differs from the original, as sessions completed after the first generation will change the list. The disclaimer states plainly that Not Currently Eligible is not an admission decision, not a refusal and not a statement about any other qualification, that More Information Needed means only that the session was incomplete, and that Grade 11 marks are projected and will change.

6.3.2 Report Layout



Learners Not Currently Eligible FUTURE PATH

Exception report · advisory sessions requiring follow-up

Period: 1 January – 15 September 2026 Generated by: A. Naidoo, System Administrator Generated: 15 September 2026, 14:32 Exceptions listed: 35 of 128 Report FP-R3

Sessions in scope: 128 Exceptions: 35 (27.3%) Not currently eligible: 29 More information needed: 6 Learners affected: 33 Access: Advisors and Administrators only

WHERE TO START 39 reasons recorded across 35 exceptions

Why learners are on this list

Mathematics mark below the minimum	9
Mathematics not taken	7
Advisory session incomplete	6
Physical Sciences not taken	5
Physical Sciences below the minimum	4
Another subject not taken	4
Another mark below the minimum	4

The three leading reasons cover 56.4% of this list, and all three are Mathematics or completeness issues that can still be fixed before final results.

Which faculties carry the follow-up

Health Sciences	9	25.7%
Science	9	25.7%
Engineering & Built Env.	7	20.0%
Commerce	6	17.1%
Humanities	4	11.4%

Health Sciences and Science together account for just over half the list. No exception in this period is older than 21 days, so the whole list is still current.

EXCEPTION LISTING Faculty, then outcome, then session date (newest first)

Session	Date	Learner	Gr	School	Qualification selected	Outcome	Reason for the outcome	Alternative
Faculty of Commerce 6 exceptions · 5 not currently eligible · 1 more information needed								
2026-0389	12 Sep 2026	Thandeka Mokoena	12	Rustenburg Secondary	BCom Accounting	✕ Not currently eligible	Mathematics 46% < 50% minimum	BCom Bus.
2026-0371	10 Sep 2026	Junaid Abrahams	12	Belhar High School	BCom Information Systems	✕ Not currently eligible	Mathematics not taken: Maths Lit held	BAdmin
2026-0344	08 Sep 2026	Naledi Sithole	11	Soweto Secondary	BCom Accounting	✕ Not currently eligible	Accounting not taken	BCom IS
2026-0318	05 Sep 2026	Chad Petersen	12	Paarl Boys' High	BCom Information Systems	✕ Not currently eligible	Mathematics 48% < 50% minimum	BSocSc
2026-0290	02 Sep 2026	Zinhle Ndlovu	12	Durban Girls' Sec.	BCom Accounting	✕ Not currently eligible	Mathematics and Accounting not taken	BA Law
2026-0265	29 Aug 2026	Ayanda Khumalo	11	Mthatha High School	BCom Information Systems	– More information needed	4 of 7 required subjects captured	–
Faculty of Science 9 exceptions · 7 not currently eligible · 2 more information needed								
2026-0412	14 Sep 2026	Aadiel-Raboo Magwa	12	Athlone High School	BSc Computer Science	✕ Not currently eligible	Mathematics not taken: Maths Lit held	BCom IS
2026-0398	13 Sep 2026	Tebogo Mahlangu	12	Mamelodi Secondary	BSc Computer Science	✕ Not currently eligible	Mathematics 52% < 65% minimum	BCom IS
2026-0377	11 Sep 2026	Kelebogile Moloi	12	Klerksdorp High	BSc Mathematical Statistics	✕ Not currently eligible	Mathematics 55% < 70% minimum	BCom Acc.
2026-0352	09 Sep 2026	Shaun Fortuin	11	Mitchells Plain Sec.	BSc Biological Sciences	✕ Not currently eligible	Life Sciences not taken	BSc EGS
2026-0331	07 Sep 2026	Nomvula Dlamini	12	PMB Girls' High	BSc Computer Science	✕ Not currently eligible	Mathematics 61% < 65% minimum	BSc Stats
2026-0307	04 Sep 2026	Farhaan Ismail	12	Lenasia Secondary	BSc Biological Sciences	✕ Not currently eligible	Life Sciences 54% < 60% minimum	BSc EGS
2026-0281	01 Sep 2026	Boitumelo Rampa	11	Polokwane High	BSc Mathematical Statistics	✕ Not currently eligible	Mathematics not taken	BA Law
2026-0259	28 Aug 2026	Lisakhanya Gxoyiya	12	Khayelitsha Sec.	BSc Computer Science	– More information needed	Interest questionnaire not completed	–
2026-0243	26 Aug 2026	Michaela Daniels	11	Kimberley Girls' High	BSc Biological Sciences	– More information needed	5 of 7 required subjects captured	–

Listing continues on page 2 with Engineering and the Built Environment, Health Sciences and Humanities.

Exception listing for advisory follow-up not an admission decision and not a refusal. Every learner listed may still apply.15 Sep 2026, 14:32 Page 1 of 2



Learners Not Currently Eligible Exception listing continued · 1 Jan – 15 Sep 2026 FUTURE PATH

EXCEPTION LISTING CONTINUED Faculty, then outcome, then session date (newest first)

Session	Date	Learner	Gr	School	Qualification selected	Outcome	Reason for the outcome	Alternative
Faculty of Engineering and the Built Environment 7 exceptions · 6 not currently eligible · 1 more information needed								
2026-0405	14 Sep 2026	Sibusiso Nkosi	12	Vereeniging Tech High	BSc Eng (Mechanical)	✕ Not currently eligible	Physical Sciences 58% < 65% minimum	BSc HLS
2026-0384	12 Sep 2026	Reece van Wyk	12	Bellville Tech High	BSc Eng (Electrical)	✕ Not currently eligible	Mathematics 60% < 70% minimum	BSc CS
2026-0360	10 Sep 2026	Karabo Lekota	12	Witbank High School	BSc Eng (Mechanical)	✕ Not currently eligible	Physical Sciences not taken	BCom IS
2026-0336	08 Sep 2026	Anele Mbatha	11	Newcastle Secondary	BSc Eng (Electrical)	✕ Not currently eligible	Mathematics + Phys. Sci. not taken	BA Law
2026-0312	05 Sep 2026	Devon Jantjies	12	Upington High School	BSc Eng (Mechanical)	✕ Not currently eligible	Mathematics 64% < 70% minimum	BSc Stats
2026-0288	02 Sep 2026	Tumi Maseko	12	Rustenburg Girls' High	BSc Eng (Electrical)	✕ Not currently eligible	Physical Sciences 61% < 65% minimum	BSc CS
2026-0261	29 Aug 2026	Liam Botha	11	George High School	BSc Eng (Mechanical)	– More information needed	Preferred faculty not selected	–
Faculty of Health Sciences 9 exceptions · 8 not currently eligible · 1 more information needed								
2026-0409	14 Sep 2026	Derek Vin Diesel	12	Sea Point High School	Bachelor of Health Sciences	✕ Not currently eligible	Mathematics + Phys. Sci. not taken	BA Law
2026-0391	13 Sep 2026	Palesa Motaung	12	Bloemfontein Sec.	MBChB	✕ Not currently eligible	Physical Sciences 63% < 70% minimum	BSc HLS
2026-0368	11 Sep 2026	Yusuf Patel	12	Laudium Secondary	MBChB	✕ Not currently eligible	Mathematics 66% < 70% minimum	BSc HLS
2026-0341	08 Sep 2026	Chanel Adams	11	Wynberg Girls' High	Bachelor of Health Sciences	✕ Not currently eligible	Physical Sciences not taken	BSc Bio
2026-0322	06 Sep 2026	Sipho Zulu	12	Empangeni High School	MBChB	✕ Not currently eligible	Mathematics + Phys. Sci. not taken	BSocSc
2026-0299	03 Sep 2026	Refilwe Mashaba	12	Nelspruit Secondary	Bachelor of Health Sciences	✕ Not currently eligible	Life Sciences 57% < 60% minimum	BSc EGS
2026-0274	31 Aug 2026	Kyle Solomons	12	Gqeberha High School	MBChB	✕ Not currently eligible	Physical Sciences 66% < 70% minimum	BSc HLS
2026-0251	27 Aug 2026	Nandi Mthembu	11	Ladysmith Secondary	Bachelor of Health Sciences	✕ Not currently eligible	Mathematics 49% < 55% minimum	BSc Bio
2026-0238	25 Aug 2026	Tasneem Jacobs	12	Rylands High School	MBChB	– More information needed	Interest questionnaire not completed	–
Faculty of Humanities 4 exceptions · 3 not currently eligible · 1 more information needed								
2026-0401	13 Sep 2026	Owethu Mpofu	12	Alexandra Secondary	BA Law	✕ Not currently eligible	English 42% < 60% minimum	BSocSc
2026-0357	09 Sep 2026	Jarrod Meyer	11	Somerset West High	BSocSc Politics & Gov.	✕ Not currently eligible	English 51% < 55% minimum	BAdmin
2026-0296	03 Sep 2026	Lindiwe Nyathi	12	Thohoyandou Sec.	BSocSc Politics & Gov.	✕ Not currently eligible	History not taken	BA Law
2026-0247	26 Aug 2026	Ruan Steyn	12	Middelburg High	BA Law	– More information needed	6 of 7 required subjects captured	–
				Report total all 5 faculties		**35 exceptions**	**29 not currently eligible**	**6 more info**

Advisory disclaimer

Not Currently Eligible means the learner did not meet one or more stored requirements for the qualification selected on that date. It is **not** an admission decision, a refusal, or a statement about any other qualification. **More Information Needed** means the session was incomplete and no check was run. Grade 11 marks are projected and will change. This report contains learner and school data and is restricted to Career Advisors and System Administrators.

Exception listing for advisory follow-up not an admission decision and not a refusal. Every learner listed may still apply.15 Sep 2026, 14:32 Page 2 of 2

7. INPUT-OUTPUT STANDARDS & CONTROLS

Future Path's advisory outcomes are only as trustworthy as the data behind them. Because the system relies entirely on locally captured learner data and locally stored reference data, with no external source to cross-check against. This section identifies the controls that keep that data accurate, valid and appropriate at every point it is captured, processed or presented.

FORMALISED OUTPUTS:

Data captured	**Format / entry convention**
Learner grade	Selected from a fixed list (Grade 11 / Grade 12), never free text
School subjects	Selected from the stored Subject lookup list, not typed in freely, so a learner cannot enter a subject the school system does not recognise
Marks	Numeric, whole or one-decimal percentage, entered per subject
Faculty / degree programme	Selected from stored Faculty and Degree Programme lookup lists
Interest questionnaire	Fixed set of stored questions, answered on a structured response scale rather than free text, so responses can be scored consistently
Funding preference	Simple yes/no selection to view funding options
Learner identifying details (name, school, province, contact)	Optional free-text fields, kept to the minimum needed for an advisory session in line with the system's privacy stance

BUILT-IN VALIDATION

Validation is applied at two points for every rule below, once in the presentation layer (so the learner or staff member gets immediate feedback) and again in the business logic layer (so a rule can never be bypassed by a malformed or unexpected request from the interface layer):

- Marks must fall between 0 and 100.
- Required fields (e.g. grade, at least one subject and mark, a selected faculty or degree) may not be left blank before the learner can proceed.
- The same subject may not be captured twice within one advisory session, the business logic layer checks existing Learner Subject entries for the current session before allowing a new one to be added.
- Dates (such as a funding option's closing date) must be valid calendar dates, entered through a date picker control rather than free text.
- A degree programme must be linked to a valid, existing faculty and enforced by the foreign key between Degree Programme and Faculty, and by only offering existing faculties in the degree-maintenance form.
- Funding type may only be selected from the predefined set of values (e.g. Scholarship, Bursary, Financial Aid, External Funding), never entered as free text.

INPUT INTEGRITY CONTROLS

- **Value Limit Controls:** Numeric fields such as marks are constrained to their valid range (0–100), checking the range of inputs for reasonableness.
- **Data Validation Controls:** Dates, numeric fields and contact details are validated against their expected format before being accepted; subjects, faculties, degrees, interest categories and funding types are selected from stored reference data (lookup lists) rather than typed. Together these ensure that specific data values entered are correct and that inconsistent or invalid values cannot enter the database.
- **Completeness Controls:** The interface will not allow a learner to proceed to the eligibility check, or an administrator to save a new record, while mandatory fields are empty — ensuring all required data has been entered.
- **Field Combination Controls:** The business logic layer rejects a second entry for a subject already captured in the same advisory session, ensuring data is correct based on the relationship between the learner, subject and advisory session records.

- **Referential integrity:** Foreign key relationships (Degree Programme → Faculty, Programme Requirement → Degree Programme, Learner Subject → Subject, Funding Faculty / Funding Programme → Funding Option, and so on) are enforced at the database level, so a record can never reference a faculty, degree or subject that does not exist.
- **Confirmation messages:** Actions that change stored data meaningfully (submitting an advisory session, exporting a report, or an administrator deactivating a degree or funding option) prompt the user to confirm before the action is finalised.

OUTPUT INTEGRITY CONTROLS

Output controls ensure that output arrives at the proper destination (for authorised eyes) and is accurate, current and complete:

- **Accuracy**: Every advisory outcome (Strong Match, Possible Match, Not Currently Eligible, More Information Needed) is displayed together with the specific reason for that outcome for example, which subject or mark requirement was not met, rather than the outcome label alone.
- **Currency**: Recommendation and reporting queries exclude degree programmes and funding options that an administrator has marked inactive, so learners are never advised toward outdated options.
- **Proper destination** (authorised eyes): Reports are only accessible to the roles they are intended for, the learner advisory summary is available to the learner who generated it, while the aggregate eligibility report and the exception report are only available to Career Advisors and System Administrators. This is enforced by both the menu options a role can see and a check in the business logic layer before a report is generated.
- **Completeness and source labelling**: A standard advisory disclaimer is shown on-screen and included on every exported report, making clear that Future Path provides preliminary guidance and not an official admission decision. Every exported report (the learner advisory summary and the two advisor/administrator reports) displays the date it was generated. Funding and programme requirement information is clearly marked as locally stored sample data, with a note that details should be verified against the official source before a learner relies on them, consistent with the principle that output labels should correctly identify the source of the data.

8. SECURITY AND ACCESS DESIGN

OUTPUT INTEGRITY CONTROLS

Future Path recognizes three user roles: Learner, Career Advisor, and System Administrator. Consistent with the case study's privacy stance, a Learner can use the core advising functionality (profile, subjects, interests, eligibility check, funding view, export) without authenticating, since advisory sessions are designed to work with minimal or no identifying information. Career Advisors and System Administrators, on the hand must log in with a valid, active User Account record before any advisor- or administrator-specific functionality becomes available.

OUTPUT INTEGRITY CONTROLS

Once a Career Advisor or Administrator logs in, the system reads the Role value on their User Account record and uses it to determine what they can do:

- **Career Advisors** can view learner advisory sessions and outcomes and generate the two advisor-facing reports (the Degree Interest and Eligibility Report and the Learners Not Currently Eligible Exception Report). They cannot create, edit or deactivate faculties, degree programmes, programme requirements, interest questionnaire content or funding options.
- **System Administrators** have access to everything a Career Advisor has, plus the reference-data maintenance functionality: adding, updating or deactivating faculties, degree programmes, programme requirements, interest categories/questions and funding options, as well as maintaining user accounts.

Capability	**Learner**	**Career Advisor**	**System Administrator**
Complete an advisory session (subjects, interests, eligibility check, funding view, export)	Yes	_	_
View advisory sessions/outcomes across learners	_	Yes	Yes
Generate Degree Interest and Eligibility Report	_	Yes	Yes
Generate Learners Not Currently Eligible exception report	_	Yes	Yes
Maintain faculties, degrees and programme requirements	_	No	Yes
Maintain funding options	_	No	Yes
Maintain interest questionnaire content	_	No	Yes
Maintain user accounts	_	No	Yes

HOW THE INTERFACE AND DATA DESIGN SUPPORT THIS

- **Interface layer**: The main menu presented after launch differs by session type: an unauthenticated Learner sees only the advising workflow; a logged-in Career Advisor sees the advising-activity and reporting views; a logged-in Administrator additionally sees the reference-data maintenance forms. Administrator-only forms are never rendered or enabled for an Advisor session.
- **Business logic layer**: Access control is not left to the interface alone. Each method that performs a reference-data change (for example, saving a programme requirement or deactivating a funding option) independently checks that the current session's role is Administrator before executing, so the restriction holds even if a request reaches the business logic layer through an unexpected path.
- **Data layer**: the User Account entity stores each staff user's credentials, role and active status, and is the single source of truth the login process checks against. Foreign key relationships tie reference-data tables (Programme Requirement, Funding Option, Interest Question, and so on) to the entities an Administrator manages, keeping maintenance actions and the data, they affect consistent with the role model above.

9. IMPLEMENTATION PLAN.

This plan shows how we will move from the approved design to the completed Future Path system. Our implementation will follow a staged approach, with some activities taking place at the same time where possible. This allows different team members to work on separate parts of the system without waiting unnecessarily, while still making sure that important dependencies are completed in the correct order. The database structure, domain classes, Learner screens and core recommendation logic will be developed early, since these provide the foundation for the rest of the system. The eligibility and recommendation logic will be developed and tested using the domain classes and test data while the Data Layer is still being completed, and will then be connected to the stored database data during integration. The Data Layer and controller classes will also be finished by this point, ready for the three tiers to be connected. Reports will be implemented once the main system is working together, and testing will take place throughout development as well as through full system testing before submission.

Stage	Development Activity	What will be done	Responsibility	Start Date	End Date	Key Milestone
1	Create the database	Create the database tables, primary and foreign keys, and relationships shown in our final ERD.	Zubair Rayned	22 Sep 2026	24 Sep 2026	Database created and ready for data
2	Build the entity/domain classes	Create the entity classes shown in our Design Class Diagram, including classes for learners, programmes, programme requirements, advisory sessions, recommendation results and user accounts.	Nabeel Braaf	22 Sep 2026	24 Sep 2026	Entity/domain classes complete

3	Build the Learner screen layouts and controls	Build the Learner screen layouts and controls for the Learner Profile, Subjects and Marks, Interest Questionnaire, Faculty and Programme Selection, Eligibility Results, Alternative Programmes, Funding Options and Advisory Summary from our final UI designs.	Kealon Vassen	22 Sep 2026	29 Sep 2026	Learner screen layouts and controls complete
4	Add sample and test data to the database	Populate the database with the learner, subject, programme, requirement, interest, funding and advisory-session test data from Part B,together with additional realistic test data.	Zubair Rayned & Nabeel Braaf	25 Sep 2026	27 Sep 2026	Test database populated
5	Implement eligibility and recommendation logic	Implement the rules for checking subjects and marks, calculating interest alignment, identifying gaps, producing advisory outcomes and ranking alternative programmes.	Oyama Nongcula	25 Sep 2026	2 Oct 2026	Eligibility and recommendation logic working

6	Build the Data Layer classes	Create the data-access classes used to retrieve, save and update information in the local database.	Zubair Rayned & Nabeel Braaf	28 Sep 2026	2 Oct 2026	Data Layer complete and working
7	Build the Career Advisor and Administrator screen layouts and controls	Build the Staff login, Career Advisor functions, Administrator screens for managing faculties, degree programmes, programme requirements, funding information, interest questions and user accounts, and the report generation screen from our final UI designs	Kealon Vassen	30 Sep 2026	5 Oct 2026	Staff screen layouts and controls complete
8	Build the controller classes	Create the controller classes that receive requests from the screens, manage the required processing, control staff access and communicate with the Data Layer to retrieve or save information.	Oyama Nongcula	3 Oct 2026	5 Oct 2026	Controller classes complete
9	Integrate the three tiers	Connect the Presentation Layer, Business Logic Layer and Data Layer so that the screens, controllers, rules and database work together correctly.	Whole team	6 Oct 2026	9 Oct 2026	Three-tier integration complete

10	Build and connect the reports	Create and connect the Learner Advisory Summary Report, Degree Interest and Eligibility Report and Learners Not Currently Eligible Exception Report using the integrated system data.	Kealon Vassen & Zubair Rayned	10 Oct 2026	11 Oct 2026	Three required reports complete and working
Ongoing throughout development	Test during development	Test completed components as they are built, record problems, fix them and retest affected functionality.	Whole team	25 Sep 2026	11 Oct 2026	Components tested throughout development
11	Full system testing	Test the complete system using valid, invalid, boundary and exception scenarios, including the required Part B scenarios.	Whole team	12 Oct 2026	14 Oct 2026	Full system testing completed
12	Final fixes and submission checks	Fix any remaining problems found during full system testing, retest the affected functionality, and complete the final submission checks.	Whole team	15 Oct 2026	16 Oct 2026	Final system ready for submission
13	Final submission	Keep this day available for any unexpected problems, complete the final checks and submit the finished system.	Whole team	16 Oct 2026	16 Oct 2026	Final system submitted

14	Prepare for the final demonstration	Rehearse the demonstration and prepare the required test cases and sample data for the markers.	Whole team	Week of 19 Oct 2026	Week of 19 Oct 2026	Final demonstration ready

10. TEST PLAN

Testing will check that the Future Path system works as required, saves the correct data and handles invalid input correctly. We will test during development and again after the full system has been connected

TEST ENVIRONMENT

Future Path will be developed and tested using:

- Windows laptops/computers
- Microsoft Visual Studio as the development environment
- C#.NET as the programming language/framework
- Windows Forms for the user interface
- SQL Server Express LocalDB as the local relational database used to store the system data
- A three-tier architecture consisting of the Presentation Layer, Business Logic Layer and Data Layer
- A standalone offline desktop application

TEST ITEMS

The main areas that will be tested are shown below.

Item to Test	Test Description	Planned Date	Responsibility
Learner profile creation	Check that learner details can be entered and saved correctly.	During development	Whole team
Subject and mark capture	Check that subjects and marks can be entered and saved correctly.	During development	Whole team

Input Validation	Check that invalid or missing input is handled correctly.	During development	Whole team
Interest questionnaire	Check that questionnaire answers are processed correctly.	During development	Whole team
Faculty and degree selection	Check that a learner can select a valid faculty and degree programme.	During development	Whole team
Degree eligibility	Check that the correct eligibility result is produced.	During development	Whole team
Advisory outcomes	Check that the correct advisory outcome and reason are shown.	During development	Whole team
Alternative recommendations	Check that suitable alternative degrees are shown when required.	During development	Whole team
Funding information	Check that the correct funding information is displayed.	During development	Whole team
Advisory summary and export report	Check that the advisory summary is generated and can be exported.	During development	Whole team
Career Advisor functionality	Check that Career Advisor functions work correctly.	During development	Whole team

Administrator functionality	Check that Administrator functions work correctly.	During development	Whole team
Staff login and role-based access	Check login and that users can only access functions allowed for their role.	During development	Whole team
Required reports	Check that all three required reports are generated correctly.	During development	Whole team

TEST APPROACHES

Testing will be done during development and again after the main parts of the system have been connected. Each test will compare the actual result produced by the system with the expected result.

Types of tests	How it will be tested
Unit testing	Test one method, class or small part of the system by itself. Examples include a mark range check, an interest score calculation or a degree requirement check.
Integration testing	Test connected parts of the system together. For example, a Windows Form sends data to the Business Logic Layer, which then uses the Data Layer to save or retrieve data from the database.
System testing	Test the complete Future Path system from start to finish, including learner functions, Career Advisor and Administrator functions, reports and the local database.

The following types of test scenarios will also be used:

Test scenarios	**How it will be tested**
Valid tests	Use correct data that should be accepted. For example, entering a valid mark such as 75.
Invalid tests	Use data that should be rejected. For example, entering letters for a mark or entering the same subject twice.
Boundary tests	Test values at the edge of what is allowed. For example, marks of -1, 0, 100 and 101 can be tested. Marks of 0 and 100 should be accepted, while -1 and 101 should be rejected.
Exception tests	Test unusual or incomplete situations without allowing the system to crash. For example, if information required to complete an eligibility check is missing, the system should give a More Information Needed outcome where appropriate.

The supplied Part B data will be used as an important check for the eligibility and recommendation logic. The actual outcome produced by the system will be compared with the expected outcome provided in the case study.

Learner	**Selected programme**	**Expected outcome**
Gadija Jacobs	BCom Information Systems	Strong Match
Aadiel-Raboo Magwa	BSc Computer Science	Not currently eligible
Labeeqah Rademeyer	BSc Biological Sciences	Possible Match – improve Mathematics
Siyabonga Dickson	BSc Engineering (Mechanical Engineering)	Strong Match
Derek Vin Diesel	Bachelor of Health Sciences	Not currently eligible – Mathematics and Physical Sciences required

When a test fails, the problem will be written down, fixed and tested again. Related tests that passed before will also be repeated where needed to make sure that a fix has not caused a problem somewhere else in the system

PROBLEM TRACKING (TEST CASES)

The following 5 test cases will be used:

ID	Test scenario	Preconditions	Steps/User Actions	Test data	Expected System Behavior or State	Actual Result/ Comments
TC01	Create learner profile and capture subjects and marks	Future Path is open and the learner profile screen is available	1. Create a new learner profile. 2. Enter the learner's grade. 3. Capture the learner's subjects and marks. 4. Save the information.	Gadija Jacobs, Grade 12. English 72%, Mathematics 70%, Accounting 74%, Business Studies 79% Economics 70%, Life Orientation 80%.	The learner profile, subjects and marks are accepted and saved successfully. All marks are within the allowed 0-100 range.	To be completed during testing.

TC02	Check eligibility for a selected degree	The learner profile, subjects and marks have already been captured and the programme requirements exist in the database.	1. Select the learner. 2. Select BCom Information Systems. 3. Run the eligibility check	Gadija Jacobs - BCom Information Systems	The system displays Strong Match and gives a reason for the advisory outcome.	To be completed during testing.
TC03	Generate alternative recommendations	The learner's subjects, marks and interest information have been captured, and alternative degree programmes and requirements exist in the database.	1. Select the learner. 2. Select BSc Computer Science. 3. Run the eligibility check. 4. View alternative recommendations.	Aadiel-Raboo Magwa, Grade 12. English 65%, Mathematical Literacy 78%, Business Studies 71%, History 69%, Tourism 75%, Life Orientation 82%.	The system displays Not Currently Eligible, explains which requirement is not met, and displays suitable alternative programmes based on the learner's subjects, marks and interests.	To be completed during testing.

TC04	View funding information	Funding information has been stored in the database and linked to the relevant faculty or programme.	1. Select a learner and programme. 2. Choose to view funding information.	Gadija Jacobs - BCom Information Systems	The system displays active funding options linked to Commerce or the selected programme, including the relevant funding information stored in the database.	To be completed during testing.
TC05	Export learner advisory summary	The learner has completed an advisory session and an advisory outcome has been generated.	1. Open the learner's advisory summary. 2. Select the export option. 3. Save/export the report.	Labeeqah Rademeyer – BSc Biological Sciences, expected Possible Match with Mathematics identified as a gap.	The advisory summary is successfully generated and exported. It contains the learner information, subjects and marks, selected programme, interest result, eligibility outcome, identified gaps, alternatives where relevant, funding information, date generated and advisory disclaimer.	To be completed during testing.

11. RELATIONSHIP BETWEEN DESIGN AND DEVELOPMENT

The Systems Design Specification provides the detailed design that was decided for the system Future Path. The development of the system will follow the design choices that were done in the different sections of this document. The developed system will correspond with the designs for the interfaces, classes, database, sequence diagrams, reports, validation and control, and the access control that was specified in this specification.

The Interface Design Specifications discussed in Section 2 illustrate the screens that the Learner, Career Advisor and Administrator will interact with. This outlines the screens from the Homepage to Learner Profile Capture, Subjects and Marks Capture, Interest Questionnaire, Faculty & Department Selection, Eligibility Results, Alternative Programmes, Funding Options, Advisory Summary, the staff-facing Sessions Dashboard and Reference Data Management screens. This section discusses the links between these screens as well as the screen specifications such as the button positioning, colour usage, and error messages. The class design is provided in Section 4. The section discusses the boundary classes behind each screen, the controller classes responsible for implementing the business logic, and the entity and data access classes responsible for storing and accessing the data. This section also describes the Recommendation Engine that integrates the scores from the four categories into one result and its two result classes of Match Result and Advisory Outcome.

The database structure is discussed in Section 5. This discusses the entities for storing the data of the Learner, Degree Programme, Programme Requirement, Advisory Session and Recommendation Result, as well as the funding and interest entities. The database is normalised in order to avoid any kind of repetition and use of foreign keys to maintain relations such as the relation between the Degree Programme and the Programme Requirements and the relation between the learner and the subjects being captured. The sequence diagrams are discussed in Section 3. The sequence diagrams demonstrate the interaction between the classes discussed in Section 4 as they perform the two processes of assessing whether the learner qualifies for their chosen programme and generating an advisory outcome and generating a ranked list of alternative degree recommendations when the learner does not achieve a strong match.

Section 6 provides information about the report design, this includes the three types of reports that will be produced by the system: the Learner Advisory Summary Report, Degree Interest and Eligibility Report and the Learners Not Currently Eligible Report.

Validation and control are defined in Section 7. This includes ensuring that the captured data is validated and controlled, such as the use of marks within a particular range, mandatory data fields and non-duplicate subjects, and ensuring that the output is also accurate and sensible. Access control design is explained in Section 8. It defines the authentication process where the Learner, Career Advisor, and Administrator can access the functionality, and it allows the Learner the ability to finish an entire advisory session without having to log in while the Career Advisor and Administrator must log in using a User Account.